Chapter 10

Advanced Topics in Relational Databases

This chapter introduces additional topics that are of interest to the database programmer. We begin with a section on the SQL standard for authorization of access to database elements. Next, we see the SQL extension that allows for recursive programming in SQL — queries that use their own results. Then, we look at the object-relational model, and how it is implemented in the SQL standard.

The remainder of the chapter concerns "OLAP," or on-line analytic processing. OLAP refers to complex queries of a nature that causes them to take significant time to execute. Because they are so expensive, some special technology has developed to handle them efficiently. One important direction is an implementation of relations, called the "data cube," that is rather different from the conventional bag-of-tuples approach of SQL.

10.1 Security and User Authorization in SQL

SQL postulates the existence of *authorization ID's*, which are essentially user names. SQL also has a special authorization ID called `PUBLIC`, which includes any user. Authorization ID's may be granted privileges, much as they would be in the file system environment maintained by an operating system. For example, a UNIX system generally controls three kinds of privileges: read, write, and execute. That list of privileges makes sense, because the protected objects of a UNIX system are files, and these three operations characterize well the things one typically does with files. However, databases are much more complex than file systems, and the kinds of privileges used in SQL are correspondingly more complex.

In this section, we shall first learn what privileges SQL allows on database elements. We shall then see how privileges may be acquired by users (by au-

thorization ID's, that is). Finally, we shall see how privileges may be taken away.

10.1.1 Privileges

SQL defines nine types of privileges: SELECT, INSERT, DELETE, UPDATE, REFERENCES, USAGE, TRIGGER, EXECUTE, and UNDER. The first four of these apply to a relation, which may be either a base table or a view. As their names imply, they give the holder of the privilege the right to query (select from) the relation, insert into the relation, delete from the relation, and update tuples of the relation, respectively.

A SQL statement cannot be executed without the privileges appropriate to that statement; e.g., a select-from-where statement requires the SELECT privilege on every table it accesses. We shall see how the module can get those privileges shortly. SELECT, INSERT, and UPDATE may also have an associated list of attributes, for instance, `SELECT(name, addr)`. If so, then only those attributes may be seen in a selection, specified in an insertion, or changed in an update, respectively. Note that, when granted, privileges such as these will be associated with a particular relation, so it will be clear at that time to what relation attributes `name` and `addr` belong.

The REFERENCES privilege on a relation is the right to refer to that relation in an integrity constraint. These constraints may take any of the forms mentioned in Chapter 7, such as assertions, attribute- or tuple-based checks, or referential integrity constraints. The REFERENCES privilege may also have an attached list of attributes, in which case only those attributes may be referenced in a constraint. A constraint cannot be created unless the owner of the schema in which the constraint appears has the REFERENCES privilege on all data involved in the constraint.

USAGE is a privilege that applies to several kinds of schema elements other than relations and assertions (see Section 9.2.2); it is the right to use that element in one's own declarations. The TRIGGER privilege on a relation is the right to define triggers on that relation. EXECUTE is the right to execute a piece of code, such as a PSM procedure or function. Finally, UNDER is the right to create subtypes of a given type. The matter of types appears in Section 10.4.

Example 10.1: Let us consider what privileges are needed to execute the insertion statement of Fig. 6.15, which we reproduce here as Fig. 10.1. First, it is an insertion into the relation `Studio`, so we require an INSERT privilege on `Studio`. However, since the insertion specifies only the component for attribute `name`, it is acceptable to have either the privilege INSERT or the privilege `INSERT(name)` on relation `Studio`. The latter privilege allows us to insert `Studio` tuples that specify only the `name` component and leave other components to take their default value or NULL, which is what Fig. 10.1 does.

However, notice that the insertion statement of Fig. 10.1 involves two subqueries, starting at lines (2) and (5). To carry out these selections we require

Triggers and Privileges

It is a bit subtle how privileges are handled for triggers. First, if you have the TRIGGER privilege for a relation, you can attempt to create any trigger you like on that relation. However, since the condition and action portions of the trigger are likely to query and/or modify portions of the database, the trigger creator must have the necessary privileges for those actions. When someone performs an activity that awakens the trigger, they do not need the privileges that the trigger condition and action require; the trigger is executed under the privileges of its creator.

```
1) INSERT INTO Studio(name)
2)     SELECT DISTINCT studioName
3)     FROM Movies
4)     WHERE studioName NOT IN
5)         (SELECT name
6)          FROM Studio);
```

Figure 10.1: Adding new studios

the privileges needed for the subqueries. Thus, we need the SELECT privilege on both relations involved in FROM clauses: Movies and Studio. Note that just because we have the INSERT privilege on Studio doesn't mean we have the SELECT privilege on Studio, or vice versa. Since it is only particular attributes of Movies and Studio that get selected, it is sufficient to have the privilege SELECT(studioName) on Movies and the privilege SELECT(name) on Studio, or privileges that include these attributes within a list of attributes. □

10.1.2 Creating Privileges

There are two aspects to the awarding of privileges: how they are created initially, and how they are passed from user to user. We shall discuss initialization here and the transmission of privileges in Section 10.1.4.

First, SQL elements such as schemas or modules have an owner. The owner of something has all privileges associated with that thing. There are three points at which ownership is established in SQL.

1. When a schema is created, it and all the tables and other schema elements in it are owned by the user who created it. This user thus has all possible privileges on elements of the schema.

2. When a session is initiated by a CONNECT statement, there is an opportunity to indicate the user with an AUTHORIZATION clause. For instance,

the connection statement

```
CONNECT TO Starfleet-sql-server AS conn1
    AUTHORIZATION kirk;
```

would create a connection called `conn1` to a database server whose name is `Starfleet-sql-server`, on behalf of user `kirk`. Presumably, the SQL implementation would verify that the user name is valid, for example by asking for a password. It is also possible to include the password in the `AUTHORIZATION` clause, as we discussed in Section 9.2.5. That approach is somewhat insecure, since passwords are then visible to someone looking over Kirk's shoulder.

3. When a module is created, there is an option to give it an owner by using an `AUTHORIZATION` clause. For instance, a clause

```
AUTHORIZATION picard;
```

in a module-creation statement would make user `picard` the owner of the module. It is also acceptable to specify no owner for a module, in which case the module is publicly executable, but the privileges necessary for executing any operations in the module must come from some other source, such as the user associated with the connection and session during which the module is executed.

10.1.3 The Privilege-Checking Process

As we saw above, each module, schema, and session has an associated user; in SQL terms, there is an associated authorization ID for each. Any SQL operation has two parties:

1. The database elements upon which the operation is performed and
2. The agent that causes the operation.

The privileges available to the agent derive from a particular authorization ID called the *current authorization ID*. That ID is either

a) The module authorization ID, if the module that the agent is executing has an authorization ID, or

b) The session authorization ID if not.

We may execute the SQL operation only if the current authorization ID possesses all the privileges needed to carry out the operation on the database elements involved.

Example 10.2: To see the mechanics of checking privileges, let us reconsider Example 10.1. We might suppose that the referenced tables — `Movies` and `Studio` — are part of a schema called `MovieSchema`, which was created by and is owned by user `janeway`. At this point, user `janeway` has all privileges on these tables and any other elements of the schema `MovieSchema`. She may choose to grant some privileges to others by the mechanism to be described in Section 10.1.4, but let us assume none have been granted yet. There are several ways that the insertion of Example 10.1 can be executed.

1. The insertion could be executed as part of a module created by user `janeway` and containing an `AUTHORIZATION janeway` clause. The module authorization ID, if there is one, always becomes the current authorization ID. Then, the module and its SQL insertion statement have exactly the same privileges user `janeway` has, which includes all privileges on the tables `Movies` and `Studio`.

2. The insertion could be part of a module that has no owner. User `janeway` opens a connection with an `AUTHORIZATION janeway` clause in the `CONNECT` statement. Now, `janeway` is again the current authorization ID, so the insertion statement has all the privileges needed.

3. User `janeway` grants all privileges on tables `Movies` and `Studio` to user `archer`, or perhaps to the special user `PUBLIC`, which stands for "all users." Suppose the insertion statement is in a module with the clause

   ```
   AUTHORIZATION archer
   ```

   Since the current authorization ID is now `archer`, and this user has the needed privileges, the insertion is again permitted.

4. As in (3), suppose user `janeway` has given user `archer` the needed privileges. Also, suppose the insertion statement is in a module without an owner; it is executed in a session whose authorization ID was set by an `AUTHORIZATION archer` clause. The current authorization ID is thus `archer`, and that ID has the needed privileges.

□

There are several principles that are illustrated by Example 10.2. We shall summarize them below.

- The needed privileges are always available if the data is owned by the same user as the user whose ID is the current authorization ID. Scenarios (1) and (2) above illustrate this point.

- The needed privileges are available if the user whose ID is the current authorization ID has been granted those privileges by the owner of the data, or if the privileges have been granted to user `PUBLIC`. Scenarios (3) and (4) illustrate this point.

- Executing a module owned by the owner of the data, or by someone who has been granted privileges on the data, makes the needed privileges available. Of course, one needs the EXECUTE privilege on the module itself. Scenarios (1) and (3) illustrate this point.

- Executing a publicly available module during a session whose authorization ID is that of a user with the needed privileges is another way to execute the operation legally. Scenarios (2) and (4) illustrate this point.

10.1.4 Granting Privileges

So far, the only way we have seen to have privileges on a database element is to be the creator and owner of that element. SQL provides a GRANT statement to allow one user to give a privilege to another. The first user retains the privilege granted, as well; thus GRANT can be thought of as "copy a privilege."

There is one important difference between granting privileges and copying. Each privilege has an associated *grant option*. That is, one user may have a privilege like SELECT on table Movies "with grant option," while a second user may have the same privilege, but without the grant option. Then the first user may grant the privilege SELECT on Movies to a third user, and moreover that grant may be with or without the grant option. However, the second user, who does not have the grant option, may not grant the privilege SELECT on Movies to anyone else. If the third user got the privilege with the grant option, then that user may grant the privilege to a fourth user, again with or without the grant option, and so on.

A *grant statement* has the form:

GRANT <privilege list> ON <database element> TO <user list>

possibly followed by WITH GRANT OPTION.

The database element is typically a relation, either a base table or a view. If it is another kind of element, the name of the element is preceded by the type of that element, e.g., ASSERTION. The privilege list is a list of one or more privileges, e.g., SELECT or INSERT(name). Optionally, the keywords ALL PRIVILEGES may appear here, as a shorthand for all the privileges that the grantor may legally grant on the database element in question.

In order to execute this grant statement legally, the user executing it must possess the privileges granted, and these privileges must be held with the grant option. However, the grantor may hold a more general privilege (with the grant option) than the privilege granted. For instance, the privilege INSERT(name) on table Studio might be granted, while the grantor holds the more general privilege INSERT on Studio, with grant option.

Example 10.3: User janeway, who is the owner of the MovieSchema schema that contains tables

```
Movies(title, year, length, genre, studioName, producerC#)
Studio(name, address, presC#)
```

grants the INSERT and SELECT privileges on table Studio and privilege SELECT on Movies to users kirk and picard. Moreover, she includes the grant option with these privileges. The grant statements are:

```
GRANT SELECT, INSERT ON Studio TO kirk, picard
    WITH GRANT OPTION;
GRANT SELECT ON Movies TO kirk, picard
    WITH GRANT OPTION;
```

Now, picard grants to user sisko the same privileges, but without the grant option. The statements executed by picard are:

```
GRANT SELECT, INSERT ON Studio TO sisko;
GRANT SELECT ON Movies TO sisko;
```

Also, kirk grants to sisko the minimal privileges needed for the insertion of Fig. 10.1, namely SELECT and INSERT(name) on Studio and SELECT on Movies. The statements are:

```
GRANT SELECT, INSERT(name) ON Studio TO sisko;
GRANT SELECT ON Movies TO sisko;
```

Note that sisko has received the SELECT privilege on Movies and Studio from two different users. He has also received the INSERT(name) privilege on Studio twice: directly from kirk and via the generalized privilege INSERT from picard. □

10.1.5 Grant Diagrams

Because of the complex web of grants and overlapping privileges that may result from a sequence of grants, it is useful to represent grants by a graph called a *grant diagram*. A SQL system maintains a representation of this diagram to keep track of both privileges and their origins (in case a privilege is revoked; see Section 10.1.6).

The nodes of a grant diagram correspond to a user and a privilege. Note that the ability to do something (e.g., SELECT on relation R) with the grant option and the same ability without the grant option are different privileges. These two different privileges, even if they belong to the same user, must be represented by two different nodes. Likewise, a user may hold two privileges, one of which is strictly more general than the other (e.g., SELECT on R and SELECT on $R(A)$. These two privileges are also represented by two different nodes.

If user U grants privilege P to user V, and this grant was based on the fact that U holds privilege Q (Q could be P with the grant option, or it could be

some generalization of P, again with the grant option), then we draw an arc from the node for U/Q to the node for V/P. As we shall see, privileges may be lost when arcs of this graph are deleted. That is why we use separate nodes for a pair of privileges, one of which includes the other, such as a privilege with and without the grant option. If the more powerful privilege is lost, the less powerful one might still be retained.

Example 10.4: Figure 10.2 shows the grant diagram that results from the sequence of grant statements of Example 10.3. We use the convention that a * after a user-privilege combination indicates that the privilege includes the grant option. Also, ** after a user-privilege combination indicates that the privilege derives from ownership of the database element in question and was not due to a grant of the privilege from elsewhere. This distinction will prove important when we discuss revoking privileges in Section 10.1.6. A doubly starred privilege automatically includes the grant option. □

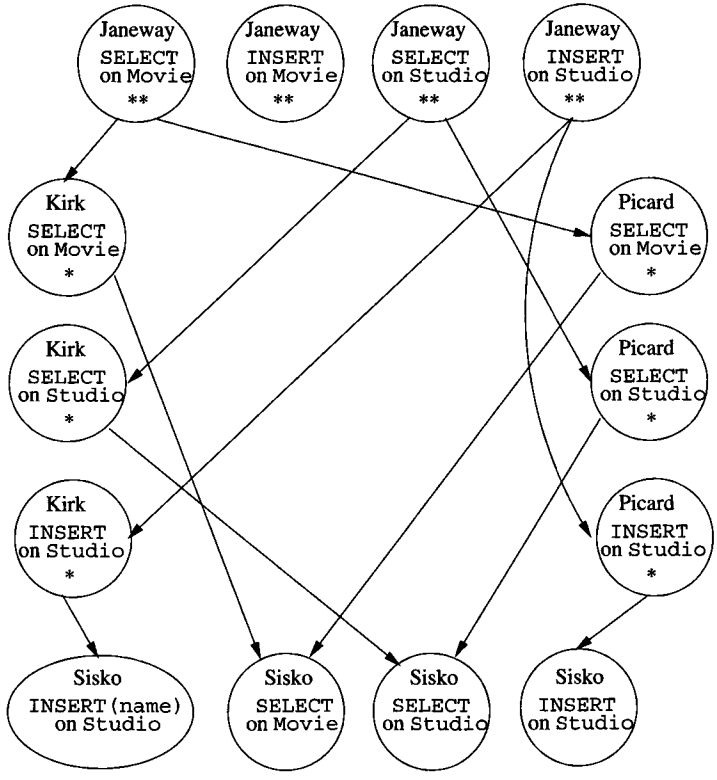


Figure 10.2: A grant diagram

10.1.6 Revoking Privileges

A granted privilege can be revoked at any time. The revoking of privileges may be required to *cascade*, in the sense that revoking a privilege with the grant option that has been passed on to other users may require those privileges to be revoked too. The simple form of a *revoke statement* begins:

`REVOKE` <privilege list> `ON` <database element> `FROM` <user list>

The statement ends with one of the following:

1. `CASCADE`. If chosen, then when the specified privileges are revoked, we also revoke any privileges that were granted *only* because of the revoked privileges. More precisely, if user U has revoked privilege P from user V, based on privilege Q belonging to U, then we delete the arc in the grant diagram from U/Q to V/P. Now, any node that is not accessible from some ownership node (doubly starred node) is also deleted.

2. `RESTRICT`. In this case, the revoke statement cannot be executed if the cascading rule described in the previous item would result in the revoking of any privileges due to the revoked privileges having been passed on to others.

It is permissible to replace `REVOKE` by `REVOKE GRANT OPTION FOR`, in which case the core privileges themselves remain, but the option to grant them to others is removed. We may have to modify a node, redirect arcs, or create a new node to reflect the changes for the affected users. This form of `REVOKE` also must be followed by either `CASCADE` or `RESTRICT`.

Example 10.5: Continuing with Example 10.3, suppose that `janeway` revokes the privileges she granted to `picard` with the statements:

```
REVOKE SELECT, INSERT ON Studio FROM picard CASCADE;
REVOKE SELECT ON Movies FROM picard CASCADE;
```

We delete the arcs of Fig. 10.2 from these `janeway` privileges to the corresponding `picard` privileges. Since `CASCADE` was stipulated, we also have to see if there are any privileges that are not reachable in the graph from a doubly starred (ownership-based) privilege. Examining Fig. 10.2, we see that `picard`'s privileges are no longer reachable from a doubly starred node (they might have been, had there been another path to a `picard` node). Also, `sisko`'s privilege to `INSERT` into `Studio` is no longer reachable. We thus delete not only `picard`'s privileges from the grant diagram, but we delete `sisko`'s `INSERT` privilege.

Note that we do not delete `sisko`'s `SELECT` privileges on `Movies` and `Studio` or his `INSERT(name)` privilege on `Studio`, because these are all reachable from `janeway`'s ownership-based privileges via `kirk`'s privileges. The resulting grant diagram is shown in Fig. 10.3. □

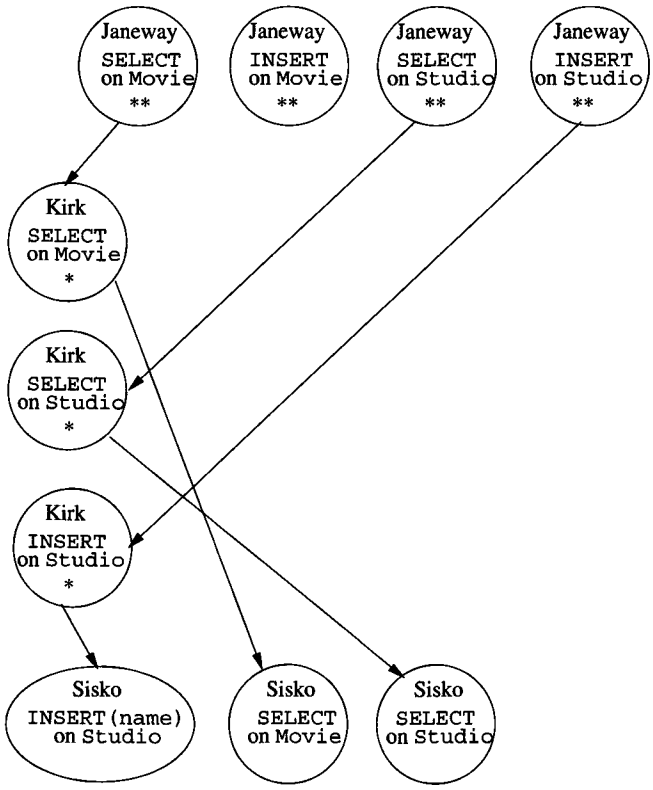


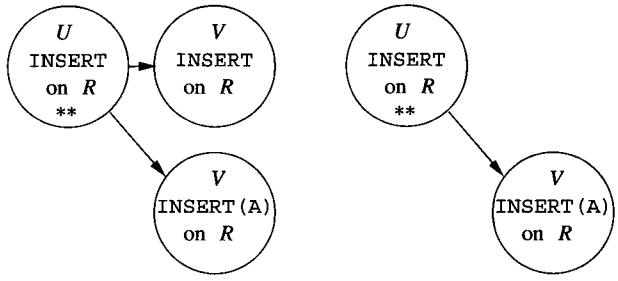
Figure 10.3: Grant diagram after revocation of picard's privileges

Example 10.6: There are a few subtleties that we shall illustrate with abstract examples. First, when we revoke a general privilege p, we do not also revoke a privilege that is a special case of p. For instance, consider the following sequence of steps, whereby user U, the owner of relation R, grants the INSERT privilege on relation R to user V, and also grants the INSERT(A) privilege on the same relation.

Step	By	Action
1	U	GRANT INSERT ON R TO V
2	U	GRANT INSERT(A) ON R TO V
3	U	REVOKE INSERT ON R FROM V RESTRICT

When U revokes INSERT from V, the INSERT(A) privilege remains. The grant diagrams after steps (2) and (3) are shown in Fig. 10.4.

Notice that after step (2) there are two separate nodes for the two similar but distinct privileges that user V has. Also observe that the RESTRICT option in step (3) does not prevent the revocation, because V had not granted the



(a) After step (2) (b) After step (3)

Figure 10.4: Revoking a general privilege leaves a more specific privilege

option to any other user. In fact, V could not have granted either privilege, because V obtained them without grant option. □

Example 10.7: Now, let us consider a similar example where U grants V a privilege $p*$ that includes the grant option and then revokes only the grant option. Assume the grant by U was based on its privilege $q*$. In this case, we must replace the arc from the $U/q*$ node to $V/p*$ by an arc from $U/q*$ to V/p, i.e., the same privilege without the grant option. If there was no such node V/p, it must be created. In normal circumstances, the node $V/p*$ becomes unreachable, and any grants of p made by V will also be unreachable. However, it may be that V was granted $p*$ by some other user besides U, in which case the $V/p*$ node remains accessible.

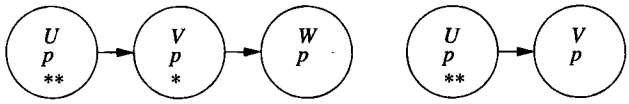
Here is a typical sequence of steps:

Step	By	Action
1	U	`GRANT` p `TO` V `WITH GRANT OPTION`
2	V	`GRANT` p `TO` W
3	U	`REVOKE GRANT OPTION FOR` p `FROM` V `CASCADE`

In step (1), U grants the privilege p to V with the grant option. In step (2), V uses the grant option to grant p to W. The diagram is then as shown in Fig. 10.5(a).

Then in step (3), U revokes the grant option for privilege p from V, but does not revoke the privilege itself. Since there is no node V/p, we create one. The arc from $U/p**$ to $V/P*$ is removed and replaced by one from $U/p**$ to V/p.

Now, the nodes $V/p*$ and W/p are not reachable from any ** node. Thus, these nodes are deleted from the diagram. The resulting grant diagram is shown in Fig. 10.5(b). □



(a) After step (2) (b) After step (3)

Figure 10.5: Revoking a grant option leaves the underlying privilege

10.1.7 Exercises for Section 10.1

Exercise 10.1.1: Indicate what privileges are needed to execute the following queries. In each case, mention the most specific privileges as well as general privileges that are sufficient.

a) The query of Fig. 6.5.

b) The query of Fig. 6.7.

c) The insertion of Fig. 6.15.

d) The deletion of Example 6.37.

e) The update of Example 6.39.

f) The tuple-based check of Fig. 7.3.

g) The assertion of Example 7.11.

Exercise 10.1.2: Show the grant diagrams after steps (4) through (6) of the sequence of actions listed in Fig. 10.6. Assume A is the owner of the relation to which privilege p refers.

Step	By	Action
1	A	GRANT p TO B WITH GRANT OPTION
2	A	GRANT p TO C
3	B	GRANT p TO D WITH GRANT OPTION
4	D	GRANT p TO B, C, E WITH GRANT OPTION
5	B	REVOKE p FROM D CASCADE
6	A	REVOKE p FROM C CASCADE

Figure 10.6: Sequence of actions for Exercise 10.1.2

Exercise 10.1.3: Show the grant diagrams after steps (5) and (6) of the sequence of actions listed in Fig. 10.7. Assume A is the owner of the relation to which privilege p refers.

Step	By	Action
1	A	GRANT p TO B, E WITH GRANT OPTION
2	B	GRANT p TO C WITH GRANT OPTION
3	C	GRANT p TO D WITH GRANT OPTION
4	E	GRANT p TO C
5	E	GRANT p TO D WITH GRANT OPTION
6	A	REVOKE GRANT OPTION FOR p FROM B CASCADE

Figure 10.7: Sequence of actions for Exercise 10.1.3

! **Exercise 10.1.4:** Show the final grant diagram after the following steps, assuming A is the owner of the relation to which privilege p refers.

Step	By	Action
1	A	GRANT p TO B WITH GRANT OPTION
2	B	GRANT p TO B WITH GRANT OPTION
3	A	REVOKE p FROM B CASCADE

10.2 Recursion in SQL

The SQL-99 standard includes provision for recursive definitions of queries. Although this feature is not part of the "core" SQL-99 standard that every DBMS is expected to implement, at least one major system — IBM's DB2 — does implement the SQL-99 proposal, which we describe in this section.

10.2.1 Defining Recursive Relations in SQL

The WITH statement in SQL allows us to define temporary relations, recursive or not. To define a recursive relation, the relation can be used within the WITH statement itself. A simple form of the WITH statement is:

WITH R AS <definition of R> <query involving R>

That is, one defines a temporary relation named R, and then uses R in some query. The temporary relation is not available outside the query that is part of the WITH statement.

More generally, one can define several relations after the WITH, separating their definitions by commas. Any of these definitions may be recursive. Several defined relations may be mutually recursive; that is, each may be defined in terms of some of the other relations, optionally including itself. However, any relation that is involved in a recursion must be preceded by the keyword RECURSIVE. Thus, a more general form of WITH statement is shown in Fig. 10.8.

Example 10.8: Many examples of the use of recursion can be found in a study of paths in a graph. Figure 10.9 shows a graph representing some flights of two

```
WITH
   [RECURSIVE] R1 AS <definition of R1>,
   [RECURSIVE] R2 AS <definition of R2>,
     ...
   [RECURSIVE] Rn AS <definition of Rn>
<query involving R1, R2, ... , Rn >
```

Figure 10.8: Form of a `WITH` statement defining several temporary relations

hypothetical airlines — *Untried Airlines* (UA), and *Arcane Airlines* (AA) — among the cities San Francisco, Denver, Dallas, Chicago, and New York. The data of the graph can be represented by a relation

```
Flights(airline, frm, to, departs, arrives)
```

and the particular tuples in this table are shown in Fig. 10.9.

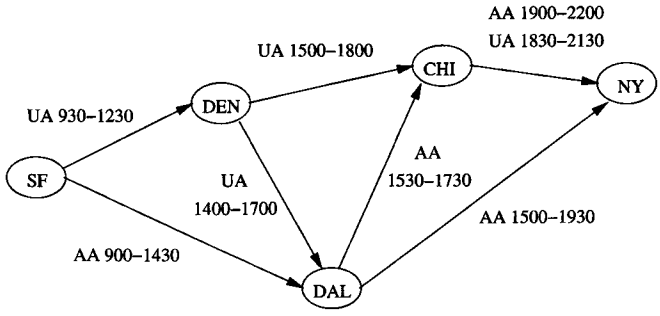


Figure 10.9: A map of some airline flights

airline	*from*	*to*	*departs*	*arrives*
UA	SF	DEN	930	1230
AA	SF	DAL	900	1430
UA	DEN	CHI	1500	1800
UA	DEN	DAL	1400	1700
AA	DAL	CHI	1530	1730
AA	DAL	NY	1500	1930
AA	CHI	NY	1900	2200
UA	CHI	NY	1830	2130

Figure 10.10: Tuples in the relation `Flights`

The simplest recursive question we can ask is "For what pairs of cities (x, y) is it possible to get from city x to city y by taking one or more flights?" Before

writing this query in recursive SQL, it is useful to express the recursion in the Datalog notation of Section 5.3. Since many concepts involving recursion are easier to express in Datalog than in SQL, you may wish to review the terminology of that section before proceeding. The following two Datalog rules describe a relation `Reaches(x,y)` that contains exactly these pairs of cities.

1. `Reaches(x,y)` ← `Flights(a,x,y,d,r)`
2. `Reaches(x,y)` ← `Reaches(x,z) AND Reaches(z,y)`

The first rule says that `Reaches` contains those pairs of cities for which there is a direct flight from the first to the second; the airline a, departure time d, and arrival time r are arbitrary in this rule. The second rule says that if you can reach from city x to city z and you can reach from z to city y, then you can reach from x to y.

Evaluating a recursive relation requires that we apply the Datalog rules repeatedly, starting by assuming there are no tuples in `Reaches`. We begin by using Rule (1) to get the following pairs in `Reaches`: (`SF`, `DEN`), (`SF`, `DAL`), (`DEN`, `CHI`), (`DEN`, `DAL`), (`DAL`, `CHI`), (`DAL`, `NY`), and (`CHI`, `NY`). These are the seven pairs represented by arcs in Fig. 10.9.

In the next round, we apply the recursive Rule (2) to put together pairs of arcs such that the head of one is the tail of the next. That gives us the additional pairs (`SF`, `CHI`), (`DEN`, `NY`), and (`SF`, `NY`). The third round combines all one- and two-arc pairs together to form paths of length up to four arcs. In this particular diagram, we get no new pairs. The relation `Reaches` thus consists of the ten pairs (x, y) such that y is reachable from x in the diagram of Fig. 10.9. Because of the way we drew the diagram, these pairs happen to be exactly those (x, y) such that y is to the right of x in Fig 10.9.

From the two Datalog rules for `Reaches` in Example 10.8, we can develop a SQL query that produces the relation `Reaches`. This SQL query places the Datalog rules for `Reaches` in a `WITH` statement, and follows it by a query. In our example, the desired result was the entire `Reaches` relation, but we could also ask some query about `Reaches`, for instance the set of cities reachable from Denver.

```
1)  WITH RECURSIVE Reaches(frm, to) AS
2)          (SELECT frm, to FROM Flights)
3)      UNION
4)          (SELECT R1.frm, R2.to
5)           FROM Reaches R1, Reaches R2
6)           WHERE R1.to = R2.frm)
7)  SELECT * FROM Reaches;
```

Figure 10.11: Recursive SQL query for pairs of reachable cities

Figure 10.11 shows how to express `Reaches` as a SQL query. Line (1) introduces the definition of `Reaches`, while the actual definition of this relation is in

Mutual Recursion

There is a graph-theoretic way to check whether two relations or predicates are mutually recursive. Construct a *dependency graph* whose nodes correspond to the relations (or predicates if we are using Datalog rules). Draw an arc from relation A to relation B if the definition of B depends directly on the definition of A. That is, if Datalog is being used, then A appears in the body of a rule with B at the head. In SQL, A would appear in a `FROM` clause, somewhere in the definition of B, possibly in a subquery. If there is a cycle involving nodes R and S, then R and S are *mutually recursive*. The most common case will be a loop from R to R, indicating that R depends recursively upon itself.

lines (2) through (6).

That definition is a union of two queries, corresponding to the two Datalog rules by which `Reaches` was defined. Line (2) is the first term of the union and corresponds to the first, or basis rule. It says that for every tuple in the `Flights` relation, the second and third components (the `frm` and `to` components) are a tuple in `Reaches`.

Lines (4) through (6) correspond to Rule (2), the recursive rule, in the definition of `Reaches`. The two `Reaches` subgoals in Rule (2) are represented in the `FROM` clause by two aliases `R1` and `R2` for `Reaches`. The first component of `R1` corresponds to x in Rule (2), and the second component of `R2` corresponds to y. Variable z is represented by both the second component of `R1` and the first component of `R2`; note that these components are equated in line (6).

Finally, line (7) describes the relation produced by the entire query. It is a copy of the `Reaches` relation. As an alternative, we could replace line (7) by a more complex query. For instance,

```
7)  SELECT to FROM Reaches WHERE frm = 'DEN';
```

would produce all those cities reachable from Denver. □

10.2.2 Problematic Expressions in Recursive SQL

The SQL standard for recursion does not allow an arbitrary collection of mutually recursive relations to be written in a `WITH` clause. There is a small matter that the standard requires only that *linear* recursion be supported. A linear recursion, in Datalog terms, is one in which no rule has more than one subgoal that is mutually recursive with the head. Notice that Rule (2) in Example 10.8 has two subgoals with predicate `Reaches` that are mutually recursive with the head (a predicate is always mutually recursive with itself; see the box on Mutual

Recursion). Thus, technically, a DBMS might refuse to execute Fig. 10.11 and yet conform to the standard.[1]

But there is a more important restriction on SQL recursions, one that, if violated leads to recursions that cannot be executed by the query processor in any meaningful way. To be a legal SQL recursion, the definition of a recursive relation R may involve only the use of a mutually recursive relation S (including R itself) if that use is "monotone" in S. A use of S is *monotone* if adding an arbitrary tuple to S might add one or more tuples to R, or it might leave R unchanged, but it can never cause any tuple to be deleted from R. The following example suggests what can happen if the monotonicity requirement is not respected.

Example 10.9: Suppose relation R is a unary (one-attribute) relation, and its only tuple is (0). R is used as an EDB relation in the following Datalog rules:

```
1. P(x) ← R(x) AND NOT Q(x)
2. Q(x) ← R(x) AND NOT P(x)
```

Informally, the two rules tell us that an element x in R is either in P or in Q but not both. Notice that P and Q are mutually recursive.

If we start out, assuming that both P and Q are empty, and apply the rules once, we find that $P = \{(0)\}$ and $Q = \{(0)\}$; that is, (0) is in both IDB relations. On the next round, we apply the rules to the new values for P and Q again, and we find that now both are empty. This cycle repeats as long as we like, but we never converge to a solution.

In fact, there are two "solutions" to the Datalog rules:

a)	$P = \{(0)\}$	$Q = \emptyset$
b)	$P = \emptyset$	$Q = \{(0)\}$

However, there is no reason to assume one over the other, and the simple iteration we suggested as a way to compute recursive relations never converges to either. Thus, we cannot answer a simple question such as "Is $P(0)$ true?"

The problem is not restricted to Datalog. The two Datalog rules of this example can be expressed in recursive SQL. Figure 10.12 shows one way of doing so. This SQL does not adhere to the standard, and no DBMS should execute it. □

The problem in Example 10.9 is that the definitions of P and Q in Fig. 10.12 are not monotone. Look at the definition of P in lines (2) through (5) for instance. P depends on Q, with which it is mutually recursive, but adding a tuple to Q can delete a tuple from P. Notice that if $R = \{(0)\}$ and Q is empty, then $P = \{(0)\}$. But if we add (0) to Q, then we delete (0) from P. Thus, the definition of P is not monotone in Q, and the SQL code of Fig. 10.12 does not meet the standard.

[1]Note, however, that we can replace either one of the uses of **Reaches** in line (5) of Fig. 10.11 by **Flights**, and thus make the recursion linear. Nonlinear recursions can frequently — although not always — be made linear in this fashion.

```
 1)  WITH
 2)      RECURSIVE P(x) AS
 3)              (SELECT * FROM R)
 4)          EXCEPT
 5)              (SELECT * FROM Q),

 6)      RECURSIVE Q(x) AS
 7)              (SELECT * FROM R)
 8)          EXCEPT
 9)              (SELECT * FROM P)

10)  SELECT * FROM P;
```

Figure 10.12: Query with nonmonotonic behavior, illegal in SQL

Example 10.10: Aggregation can also lead to nonmonotonicity. Suppose we have unary (one-attribute) relations P and Q defined by the following two conditions:

1. P is the union of Q and an EDB relation R.
2. Q has one tuple that is the sum of the members of P.

We can express these conditions by a `WITH` statement, although this statement violates the monotonicity requirement of SQL. The query shown in Fig. 10.13 asks for the value of P.

```
1)  WITH
2)      RECURSIVE P(x) AS
3)              (SELECT * FROM R)
4)          UNION
5)              (SELECT * FROM Q),

6)      RECURSIVE Q(x) AS
7)          SELECT SUM(x) FROM P

8)  SELECT * FROM P;
```

Figure 10.13: Nonmonotone query involving aggregation, illegal in SQL

Suppose that R consists of the tuples (12) and (34), and initially P and Q are both empty. Figure 10.14 summarizes the values computed in the first six rounds. Note that both relations are computed, in one round, from the values of the relations at the previous round. Thus, P is computed in the first round

Round	P	Q
1)	{(12), (34)}	{NULL}
2)	{(12), (34), NULL}	{(46)}
3)	{(12), (34), (46)}	{(46)}
4)	{(12), (34), (46)}	{(92)}
5)	{(12), (34), (92)}	{(92)}
6)	{(12), (34), (92)}	{(138)}

Figure 10.14: Iterative calculation for a nonmonotone aggregation

to be the same as R, and Q is {NULL}, since the old, empty value of P is used in line (7).

At the second round, the union of lines (3) through (5) is the set

$$R \cup \{\texttt{NULL}\} = \{(12),\ (34),\ \texttt{NULL}\}$$

so that set becomes the new value of P. The old value of P was {(12), (34)}, so on the second round $Q = \{(46)\}$. That is, 46 is the sum of 12 and 34.

At the third round, we get $P = \{(12), (34), (46)\}$ at lines (2) through (5). Using the old value of P, {(12), (34), NULL}, Q is defined by lines (6) and (7) to be {(46)} again. Remember that NULL is ignored in a sum.

At the fourth round, P has the same value, {(12), (34), (46)}, but Q gets the value {(92)}, since 12+34+46=92. Notice that Q has lost the tuple (46), although it gained the tuple (92). That is, adding the tuple (46) to P has caused a tuple (by coincidence the same tuple) to be deleted from Q. That behavior is the nonmonotonicity that SQL prohibits in recursive definitions, confirming that the query of Fig. 10.13 is illegal. In general, at the $2i$th round, P will consist of the tuples (12), (34), and $(46i - 46)$, while Q consists only of the tuple $(46i)$. □

10.2.3 Exercises for Section 10.2

Exercise 10.2.1: The relation

```
Flights(airline, frm, to, departs, arrives)
```

from Example 10.8 has arrival- and departure-time information that we did not consider. Suppose we are interested not only in whether it is possible to reach one city from another, but whether the journey has reasonable connections. That is, when using more than one flight, each flight must arrive at least an hour before the next flight departs. You may assume that no journey takes place over more than one day, so it is not necessary to worry about arrival close to midnight followed by a departure early in the morning.

a) Write this recursion in Datalog.

b) Write the recursion in SQL.

! **Exercise 10.2.2:** In Example 10.8 we used `frm` as an attribute name. Why did we not use the more obvious name `from`?

Exercise 10.2.3: Suppose we have a relation

```
SequelOf(movie, sequel)
```

that gives the immediate sequels of a movie, of which there can be more than one. We want to define a recursive relation `FollowOn` whose pairs (x, y) are movies such that y was either a sequel of x, a sequel of a sequel, or so on.

a) Write the definition of `FollowOn` as recursive Datalog rules.

b) Write the definition of `FollowOn` as a SQL recursion.

c) Write a recursive SQL query that returns the set of pairs (x, y) such that movie y is a follow-on to movie x, but is not a sequel of x.

d) Write a recursive SQL query that returns the set of pairs (x, y) meaning that y is a follow-on of x, but is neither a sequel nor a sequel of a sequel.

! e) Write a recursive SQL query that returns the set of movies x that have at least two follow-ons. Note that both could be sequels, rather than one being a sequel and the other a sequel of a sequel.

! f) Write a recursive SQL query that returns the set of pairs (x, y) such that movie y is a follow-on of x but y has at most one follow-on.

Exercise 10.2.4: Suppose we have a relation

```
Rel(class, rclass, mult)
```

that describes how one ODL class is related to other classes. Specifically, this relation has tuple (c, d, m) if there is a relation from class c to class d. This relation is multivalued if $m =$ `'multi'` and it is single-valued if $m =$ `'single'`. It is possible to view `Rel` as defining a graph whose nodes are classes and in which there is an arc from c to d labeled m if and only if (c, d, m) is a tuple of `Rel`. Write a recursive SQL query that produces the set of pairs (c, d) such that:

a) There is a path from class c to class d in the graph described above.

b) There is a path from c to d along which every arc is labeled `single`.

! c) There is a path from c to d along which at least one arc is labeled `multi`.

d) There is a path from c to d but no path along which all arcs are labeled `single`.

! e) There is a path from c to d along which arc labels alternate `single` and `multi`.

f) There are paths from c to d and from d to c along which every arc is labeled `single`.

10.3 The Object-Relational Model

The relational model and the object-oriented model typified by ODL are two important points in a spectrum of options that could underlie a DBMS. For an extended period, the relational model was dominant in the commercial DBMS world. Object-oriented DBMS's made limited inroads during the 1990's, but never succeeded in winning significant market share from the vendors of relational DBMS's. Rather, the vendors of relational systems have moved to incorporate many of the ideas found in ODL or other object-oriented-database proposals. As a result, many DBMS products that used to be called "relational" are now called "object-relational."

This section extends the abstract relational model to incorporate several important object-relational ideas. It is followed by sections that cover object-relational extensions of SQL. We introduce the concept of object-relations in Section 10.3.1, then discuss one of its earliest embodiments — nested relations — in Section 10.3.2. ODL-like references for object-relations are discussed in Section 10.3.3, and in Section 10.3.4 we compare the object-relational model with the pure object-oriented approach.

10.3.1 From Relations to Object-Relations

While the relation remains the fundamental concept, the relational model has been extended to the *object-relational model* by incorporation of features such as:

1. *Structured types for attributes.* Instead of allowing only atomic types for attributes, object-relational systems support a type system like ODL's: types built from atomic types and type constructors for structs, sets, and bags, for instance. Especially important is a type that is a bag of structs, which is essentially a relation. That is, a value of one component of a tuple can be an entire relation, called a "nested relation."

2. *Methods.* These are similar to methods in ODL or any object-oriented programming system.

3. *Identifiers for tuples.* In object-relational systems, tuples play the role of objects. It therefore becomes useful in some situations for each tuple to have a unique ID that distinguishes it from other tuples, even from tuples that have the same values in all components. This ID, like the object-identifier assumed in ODL, is generally invisible to the user, although

there are even some circumstances where users can see the identifier for a tuple in an object-relational system.

4. *References.* While the pure relational model has no notion of references or pointers to tuples, object-relational systems can use these references in various ways.

In the next sections, we shall elaborate upon and illustrate each of these additional capabilities of object-relational systems.

10.3.2 Nested Relations

In the *nested-relational model*, we allow attributes of relations to have a type that is not atomic; in particular, a type can be a relation schema. As a result, there is a convenient, recursive definition of the types of attributes and the types (schemas) of relations:

BASIS: An atomic type (integer, real, string, etc.) can be the type of an attribute.

INDUCTION: A relation's type can be any *schema* consisting of names for one or more attributes, and any legal type for each attribute. In addition, a schema also can be the type of any attribute.

In what follows, we shall generally omit atomic types where they do not matter. An attribute that is a schema will be represented by the attribute name and a parenthesized list of the attributes of its schema. Since those attributes may themselves have structure, parentheses can be nested to any depth.

Example 10.11: Let us design a nested-relation schema for stars that incorporates within the relation an attribute `movies`, which will be a relation representing all the movies in which the star has appeared. The relation schema for attribute `movies` will include the title, year, and length of the movie. The relation schema for the relation `Stars` will include the name, address, and birthdate, as well as the information found in `movies`. Additionally, the `address` attribute will have a relation type with attributes `street` and `city`. We can record in this relation several addresses for the star. The schema for `Stars` can be written:

```
Stars(name, address(street, city), birthdate,
   movies(title, year, length))
```

An example of a possible relation for nested relation `Stars` is shown in Fig. 10.15. We see in this relation two tuples, one for Carrie Fisher and one for Mark Hamill. The values of components are abbreviated to conserve space, and the dashed lines separating tuples are only for convenience and have no notational significance.

name	*address*	*birthdate*	*movies*
Fisher	*street* | *city* Maple | H'wood Locust | Malibu	9/9/99	*title* | *year* | *length* Star Wars | 1977 | 124 Empire | 1980 | 127 Return | 1983 | 133
Hamill	*street* | *city* Oak | B'wood	8/8/88	*title* | *year* | *length* Star Wars | 1977 | 124 Empire | 1980 | 127 Return | 1983 | 133

Figure 10.15: A nested relation for stars and their movies

In the Carrie Fisher tuple, we see her name, an atomic value, followed by a relation for the value of the address component. That relation has two attributes, `street` and `city`, and there are two tuples, corresponding to her two houses. Next comes the birthdate, another atomic value. Finally, there is a component for the `movies` attribute; this attribute has a relation schema as its type, with components for the title, year, and length of a movie. The relation for the `movies` component of the Carrie Fisher tuple has tuples for her three best-known movies.

The second tuple, for Mark Hamill, has the same components. His relation for `address` has only one tuple, because in our imaginary data, he has only one house. His relation for `movies` looks just like Carrie Fisher's because their best-known movies happen, by coincidence, to be the same. Note that these two relations are two different tuple-components. These components happen to be identical, just like two components that happened to have the same integer value, e.g., 124. □

10.3.3 References

The fact that movies like *Star Wars* will appear in several relations that are values of the `movies` attribute in the nested relation `Stars` is a cause of redundancy. In effect, the schema of Example 10.11 has the nested-relation analog of not being in BCNF. However, decomposing this `Stars` relation will not eliminate the redundancy. Rather, we need to arrange that among all the tuples of all the `movies` relations, a movie appears only once.

To cure the problem, object-relations need the ability for one tuple t to refer to another tuple s, rather than incorporating s directly in t. We thus add to our model an additional inductive rule: the type of an attribute also can be a

reference to a tuple with a given schema or a set of references to tuples with a given schema.

If an attribute A has a type that is a reference to a single tuple with a relation schema named R, we show the attribute A in a schema as $A(*R)$. Notice that this situation is analogous to an ODL relationship A whose type is R; i.e., it connects to a single object of type R. Similarly, if an attribute A has a type that is a set of references to tuples of schema R, then A will be shown in a schema as $A(\{*R\})$. This situation resembles an ODL relationship A that has type `Set<`R`>`.

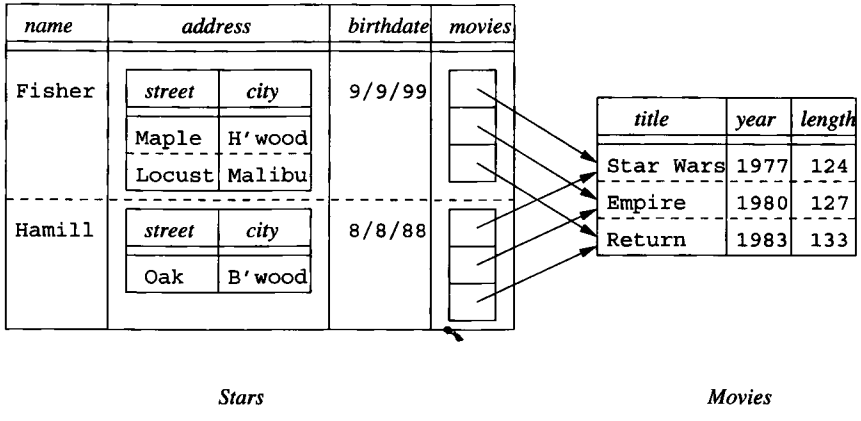


Figure 10.16: Sets of references as the value of an attribute

Example 10.12: An appropriate way to fix the redundancy in Fig. 10.15 is to use two relations, one for stars and one for movies. In this example only, we shall use a relation called `Movies` that is an ordinary relation with the same schema as the attribute `movies` in Example 10.11. A new relation `Stars` has a schema similar to the nested relation `Stars` of that example, but the movies attribute will have a type that is a set of references to `Movies` tuples. The schemas of the two relations are thus:

```
Movies(title, year, length)
Stars(name, address(street, city), birthdate,
    movies({*Movies}))
```

The data of Fig. 10.15, converted to this new schema, is shown in Fig. 10.16. Notice that, because each movie has only one tuple, although it can have many references, we have eliminated the redundancy inherent in the schema of Example 10.11. □

10.3.4 Object-Oriented Versus Object-Relational

The object-oriented data model, as typified by ODL, and the object-relational model discussed here, are remarkably similar. Some of the salient points of comparison follow.

Objects and Tuples

An object's value is really a struct with components for its attributes and relationships. It is not specified in the ODL standard how relationships are to be represented, but we may assume that an object is connected to related objects by some collection of references. A tuple is likewise a struct, but in the conventional relational model, it has components for only the attributes. Relationships would be represented by tuples in another relation, as suggested in Section 4.5.2. However the object-relational model, by allowing sets of references to be a component of tuples, also allows relationships to be incorporated directly into the tuples that represent an "object" or entity.

Methods

We did not discuss the use of methods as part of an object-relational schema. However, in practice, the SQL-99 standard and all implementations of object-relational ideas allow the same ability as ODL to declare and define methods associated with any class or type.

Type Systems

The type systems of the object-oriented and object-relational models are quite similar. Each is based on atomic types and construction of new types by struct- and collection-type-constructors. The choice of collection types may vary, but all variants include at least sets and bags. Moreover, the set (or bag) of structs type plays a special role in both models. It is the type of classes in ODL, and the type of relations in the object-relational model.

References and Object-ID's

A pure object-oriented model uses object-ID's that are completely hidden from the user, and thus cannot be seen or queried. The object-relational model allows references to be part of a type, and thus it is possible under some circumstances for the user to see their values and even remember them for future use. You may regard this situation as anything from a serious bug to a stroke of genius, depending on your point of view, but in practice it appears to make little difference.

Backwards Compatibility

With little difference in essential features of the two models, it is interesting to consider why object-relational systems have dominated the pure object-oriented systems in the marketplace. The reason, we believe, is as follows. As relational DBMS's evolved into object-relational DBMS's, the vendors were careful to maintain backwards compatibility. That is, newer versions of the system would still run the old code and accept the same schemas, should the user not care to adopt any of the object-oriented features. On the other hand, migration to a pure object-oriented DBMS would require the installations to rewrite and reorganize extensively. Thus, whatever competitive advantage could be argued for object-oriented database systems was insufficient to motivate many to make the switch.

10.3.5 Exercises for Section 10.3

Exercise 10.3.1: Using the notation developed for nested relations and relations with references, give one or more relation schemas that represent the following information. In each case, you may exercise some discretion regarding what attributes of a relation are included, but try to keep close to the attributes found in our running movie example. Also, indicate whether your schemas exhibit redundancy, and if so, what could be done to avoid it.

a) Movies, with the usual attributes plus all their stars and the usual information about the stars.

! b) Studios, all the movies made by that studio, and all the stars of each movie, including all the usual attributes of studios, movies, and stars.

c) Movies with their studio, their stars, and all the usual attributes of these.

Exercise 10.3.2: Represent the banking information of Exercise 4.1.1 in the object-relational model developed in this section. Make sure that it is easy, given the tuple for a customer, to find their account(s) and *also* easy, given the tuple for an account to find the customer(s) that hold that account. Also, try to avoid redundancy.

! **Exercise 10.3.3:** If the data of Exercise 10.3.2 were modified so that an account could be held by only one customer [as in Exercise 4.1.2(a)], how could your answer to Exercise 10.3.2 be simplified?

! **Exercise 10.3.4:** Render the players, teams, and fans of Exercise 4.1.3 in the object-relational model.

! **Exercise 10.3.5:** Render the genealogy of Exercise 4.1.6 in the object-relational model.

10.4 User-Defined Types in SQL

We now turn to the way SQL-99 incorporates many of the object-oriented features that we saw in Section 10.3. The central extension that turns the relational model into the object-relational model in SQL is the *user-defined type*, or UDT. We find UDT's used in two distinct ways:

1. A UDT can be the type of a table.
2. A UDT can be the type of an attribute belonging to some table.

10.4.1 Defining Types in SQL

The SQL-99 standard allows the programmer to define UDT's in several ways. The simplest is as a renaming of an existing type.

CREATE TYPE T AS <primitive type>;

renames a primitive type such as INTEGER. Its purpose is to prevent errors caused by accidental coercions among values that logically should not be compared or interchanged, even though they have the same primitive data type. An example should make the purpose clear.

Example 10.13: In our running movies example, there are several attributes of type INTEGER. These include `length` of `Movies`, `cert#` of `MovieExec`, and `presC#` of `Studio`. It makes sense to compare a value of `cert#` with a value of `presC#`, and we could even take a value from one of these two attributes and store it in a tuple as the value of the other attribute. However, It would not make sense to compare a movie length with the certificate number of a movie executive, or to take a `length` value from a `Movies` tuple and store it in the `cert#` attribute of a `MovieExec` tuple.

If we create types:

```
CREATE TYPE CertType AS INTEGER;
CREATE TYPE LengthType AS INTEGER;
```

then we can declare `cert#` and `presC#` to be of type `CertType` instead of INTEGER in their respective relation declarations, and we can declare `length` to be of type `LengthType` in the `Movies` declaration. In that case, an object-relational DBMS will intercept attempts to compare values of one type with the other, or to use a value of one type in place of the other. □

A more powerful form of UDT declaration in SQL is similar to a class declaration in ODL, with some distinctions. First, key declarations for a relation with a user-defined type are part of the table definition, not the type definition; that is, many SQL relations can be declared to have the same UDT but different keys and other constraints. Second, in SQL we do not treat relationships as properties. A relationship can be represented by a separate relation,

as was discussed in Section 4.10.5, or through references, which are covered in Section 10.4.5. This form of UDT definition is:

CREATE TYPE T AS (<attribute declarations>);

Example 10.14: Figure 10.17 shows two UDT's, `AddressType` and `StarType`. A tuple of type `AddressType` has two components, whose attributes are `street` and `city`. The types of these components are character strings of length 50 and 20, respectively. A tuple of type `StarType` also has two components. The first is attribute `name`, whose type is a 30-character string, and the second is `address`, whose type is itself a UDT `AddressType`, that is, a tuple with `street` and `city` components. □

```
CREATE TYPE AddressType AS (
    street  CHAR(50),
    city    CHAR(20)
);

CREATE TYPE StarType AS (
    name    CHAR(30),
    address AddressType
);
```

Figure 10.17: Two type definitions

10.4.2 Method Declarations in UDT's

The declaration of a method resembles the way a function in PSM is introduced; see Section 9.4.1. There is no analog of PSM procedures as methods. That is, every method returns a value of some type. While function declarations and definitions in PSM are combined, a method needs both a declaration, which follows the parenthesized list of attributes in the `CREATE TYPE` statement, and a separate definition, in a `CREATE METHOD` statement. The actual code for the method need not be PSM, although it could be. For example, the method body could be Java with JDBC used to access the database.

A method declaration looks like a PSM function declaration, with the keyword `METHOD` replacing `CREATE FUNCTION`. However, SQL methods typically have no arguments; they are applied to rows, just as ODL methods are applied to objects. In the definition of the method, `SELF` refers to this tuple, if necessary.

Example 10.15: Let us extend the definition of the type `AddressType` of Fig. 10.17 with a method `houseNumber` that extracts from the `street` component the portion devoted to the house address. For instance, if the `street`

component were '123 Maple St.', then houseNumber should return '123'. Exactly how houseNumber works is not visible in its declaration; the details are left for the definition. The revised type definition is thus shown in Fig. 10.18.

```
CREATE TYPE AddressType AS (
    street  CHAR(50),
    city    CHAR(20)
    )
    METHOD houseNumber() RETURNS CHAR(10);
```

Figure 10.18: Adding a method declaration to a UDT

We see the keyword METHOD, followed by the name of the method and a parenthesized list of its arguments and their types. In this case, there are no arguments, but the parentheses are still needed. Had there been arguments, they would have appeared, followed by their types, such as (a INT, b CHAR(5)). □

10.4.3 Method Definitions

Separately, we need to define the method. A simple form of method definition is:

```
CREATE METHOD <method name, arguments, and return type>
  FOR <UDT name>
    <method body>
```

That is, the UDT for which the method is defined is indicated in a FOR clause. The method definition need not be contiguous to, or part of, the definition of the type to which it belongs.

Example 10.16: For instance, we could define the method houseNumber from Example 10.15 as in Fig. 10.19. We have omitted the body of the method because accomplishing the intended separation of the string string as intended is nontrivial, even if a general-purpose host language is used. □

```
CREATE METHOD houseNumber() RETURNS CHAR(10)
FOR AddressType
BEGIN
    ...
END;
```

Figure 10.19: Defining a method

10.4.4 Declaring Relations with a UDT

Having declared a type, we may declare one or more relations whose tuples are of that type. The form of relation declarations is like that of Section 2.3.3, but the attribute declarations are omitted from the parenthesized list of elements, and replaced by a clause with `OF` and the name of the UDT. That is, the alternative form of a `CREATE TABLE` statement, using a UDT, is:

```
CREATE TABLE <table name> OF <UDT name>
    (<list of elements>);
```

The parenthesized list of elements can include keys, foreign keys, and tuple-based constraints. Note that all these elements are declared for a particular table, not for the UDT. Thus, there can be several tables with the same UDT as their row type, and these tables can have different constraints, and even different keys. If there are no constraints or key declarations desired for the table, then the parentheses are not needed.

Example 10.17: We could declare `MovieStar` to be a relation whose tuples are of type `StarType` by

```
CREATE TABLE MovieStar OF StarType (
    PRIMARY KEY (name)
);
```

As a result, table `MovieStar` has two attributes, `name` and `address`. The first attribute, `name`, is an ordinary character string, but the second, `address`, has a type that is itself a UDT, namely the type `AddressType`. Attribute `name` is a key for this relation, so it is not possible to have two tuples with the same name. □

10.4.5 References

The effect of object identity in object-oriented languages is obtained in SQL through the notion of a *reference*. A table may have a *reference column* that serves as the "identity" for its tuples. This column could be the primary key of the table, if there is one, or it could be a column whose values are generated and maintained unique by the DBMS, for example. We shall defer to Section 10.4.6 the matter of defining reference columns until we first see how reference types are used.

To refer to the tuples of a table with a reference column, an attribute may have as its type a reference to another type. If T is a UDT, then `REF`(T) is the type of a reference to a tuple of type T. Further, the reference may be given a *scope*, which is the name of the relation whose tuples are referred to. Thus, an attribute A whose values are references to tuples in relation R, where R is a table whose type is the UDT T, would be declared by:

```
CREATE TYPE StarType AS (
    name      CHAR(30),
    address   AddressType,
    bestMovie REF(MovieType) SCOPE Movies
);
```

Figure 10.20: Adding a best movie reference to StarType

A REF(T) SCOPE R

If no scope is specified, the reference can go to any relation of type T.

Example 10.18: Let us record in MovieStar the best movie for each star. Assume that we have declared an appropriate relation Movies, and that the type of this relation is the UDT MovieType; we shall define both MovieType and Movies later, in Fig. 10.21. Figure 10.20 is a new definition of StarType that includes an attribute bestMovies that is a reference to a movie. Now, if relation MovieStar is defined to have the UDT of Fig. 10.20, then each star tuple will have a component that refers to a Movies tuple — the star's best movie. □

10.4.6 Creating Object ID's for Tables

In order to refer to rows of a table, such as Movies in Example 10.18, that table needs to have an "object-ID" for its tuples. Such a table is said to be *referenceable*. In a CREATE TABLE statement where the type of the table is a UDT (as in Section 10.4.4), we may include an element of the form:

REF IS <attribute name> <how generated>

The attribute name is a name given to the column that will serve as the object-ID for tuples. The "how generated" clause can be:

1. SYSTEM GENERATED, meaning that the DBMS is responsible for maintaining a unique value in this column of each tuple, or
2. DERIVED, meaning that the DBMS will use the primary key of the relation to produce unique values for this column.

Example 10.19: Figure 10.21 shows how the UDT MovieType and relation Movies could be declared so that Movies is referenceable. The UDT is declared in lines (1) through (4). Then the relation Movies is defined to have this type in lines (5) through (7). Notice that we have declared title and year, together, to be the key for relation Movies in line (7).

We see in line (6) that the name of the "identity" column for Movies is movieID. This attribute, which automatically becomes a fourth attribute of

```
1)  CREATE TYPE MovieType AS (
2)      title   CHAR(30),
3)      year    INTEGER,
4)      genre   CHAR(10)
    );

5)  CREATE TABLE Movies OF MovieType (
6)      REF IS movieID SYSTEM GENERATED,
7)      PRIMARY KEY (title, year)
    );
```

Figure 10.21: Creating a referenceable table

`Movies`, along with `title`, `year`, and `genre`, may be used in queries like any other attribute of `Movies`.

Line (6) also says that the DBMS is responsible for generating the value of `movieID` each time a new tuple is inserted into `Movies`. Had we replaced `SYSTEM GENERATED` by `DERIVED`, then new tuples would get their value of `movieID` by some calculation, performed by the system, on the values of the primary-key attributes `title` and `year` taken from the new tuple. □

Example 10.20: Now, let us see how to represent the many-many relationship between movies and stars using references. Previously, we represented this relationship by a relation like `StarsIn` that contains tuples with the keys of `Movies` and `MovieStar`. As an alternative, we may define `StarsIn` to have references to tuples from these two relations.

First, we need to redefine $\widehat{\text{MovieStar}}$ so it is a referenceable table, thusly:

```
CREATE TABLE MovieStar OF StarType (
    REF IS starID SYSTEM GENERATED,
    PRIMARY KEY (name)
);
```

Then, we may declare the relation `StarsIn` to have two attributes, which are references, one to a movie tuple and one to a star tuple. Here is a direct definition of this relation:

```
CREATE TABLE StarsIn (
    star    REF(StarType) SCOPE MovieStar,
    movie   REF(MovieType) SCOPE Movies
);
```

Optionally, we could have defined a UDT as above, and then declared `StarsIn` to be a table of that type. □

10.4.7 Exercises for Section 10.4

Exercise 10.4.1: For our running movies example, choose type names for the attributes of each of the relations. Give attributes the same UDT if their values can reasonably be compared or exchanged, and give them different UDT's if they should not have their values compared or exchanged.

Exercise 10.4.2: Write type declarations for the following types:

a) `NameType`, with components for first, middle, and last names and a title.

b) `PersonType`, with a name of the person and references to the persons that are their mother and father. You must use the type from part (a) in your declaration.

c) `MarriageType`, with the date of the marriage and references to the husband and wife.

Exercise 10.4.3: Redesign our running products database schema of Exercise 2.4.1 to use type declarations and reference attributes where appropriate. In particular, in the relations `PC`, `Laptop`, and `Printer` make the `model` attribute be a reference to the `Product` tuple for that model.

! **Exercise 10.4.4:** In Exercise 10.4.3 we suggested that model numbers in the tables `PC`, `Laptop`, and `Printer` could be references to tuples of the `Product` table. Is it also possible to make the `model` attribute in `Product` a reference to the tuple in the relation for that type of product? Why or why not?

Exercise 10.4.5: Redesign our running battleships database schema of Exercise 2.4.3 to use type declarations and reference attributes where appropriate. Look for many-one relationships and try to represent them using an attribute with a reference type.

10.5 Operations on Object-Relational Data

All appropriate SQL operations from previous chapters apply to tables that are declared with a UDT or that have attributes whose type is a UDT. There are also some entirely new operations we can use, such as reference-following. However, some familiar operations, especially those that access or modify columns whose type is a UDT, involve new syntax.

10.5.1 Following References

Suppose x is a value of type `REF(`T`)`. Then x refers to some tuple t of type T. We can obtain tuple t itself, or components of t, by two means:

1. Operator -> has essentially the same meaning as this operator does in C. That is, if x is a reference to a tuple t, and a is an attribute of t, then `x->a` is the value of the attribute a in tuple t.

2. The `DEREF` operator applies to a reference and produces the tuple referenced.

Example 10.21: Let us use the relation `StarsIn` from Example 10.20 to find the movies in which Brad Pitt starred. Recall that the schema is

```
StarsIn(star, movie)
```

where `star` and `movie` are references to tuples of `MovieStar` and `Movies`, respectively. A possible query is:

```
1)  SELECT DEREF(movie)
2)  FROM StarsIn
3)  WHERE star->name = 'Brad Pitt';
```

In line (3), the expression `star->name` produces the value of the `name` component of the `MovieStar` tuple referred to by the `star` component of any given `StarsIn` tuple. Thus, the `WHERE` clause identifies those `StarsIn` tuples whose `star` components are references to the Brad-Pitt `MovieStar` tuple. Line (1) then produces the movie tuple referred to by the `movie` component of those tuples. All three attributes — `title`, `year`, and `genre` — will appear in the printed result.

Note that we could have replaced line (1) by:

```
1)  SELECT  movie
```

However, had we done so, we would have gotten a list of system-generated gibberish that serves as the internal unique identifiers for certain movie tuples. We would not see the information in the referenced tuples. □

10.5.2 Accessing Components of Tuples with a UDT

When we define a relation to have a UDT, the tuples must be thought of as single objects, rather than lists with components corresponding to the attributes of the UDT. As a case in point, consider the relation `Movies` declared in Fig. 10.21. This relation has UDT `MovieType`, which has three attributes: `title`, `year`, and `genre`. However, a tuple t in `Movies` has only *one* component, not three. That component is the object itself.

If we "drill down" into the object, we can extract the values of the three attributes in the type `MovieType`, as well as use any methods defined for that type. However, we have to access these attributes properly, since they are not attributes of the tuple itself. Rather, every UDT has an implicitly defined *observer method* for each attribute of that UDT. The name of the observer

method for an attribute x is $x()$. We apply this method as we would any other method for this UDT; we attach it with a dot to an expression that evaluates to an object of this type. Thus, if t is a variable whose value is of type T, and x is an attribute of T, then $t.x()$ is the value of x in the tuple (object) denoted by t.

Example 10.22: Let us find, from the relation `Movies` of Fig. 10.21 the year(s) of movies with title *King Kong*. Here is one way to do so:

```
SELECT m.year()
FROM Movies m
WHERE m.title() = 'King Kong';
```

Even though the tuple variable m would appear not to be needed here, we need a variable whose value is an object of type `MovieType` — the UDT for relation `Movies`. The condition of the `WHERE` clause compares the constant `'King Kong'` to the value of `m.title()`, the observer method for attribute `title` applied to a `MovieType` object m. Similarly, the value in the `SELECT` clause is expressed `m.year()`; this expression applies the observer method for `year` to the object m. □

In practice, object-relational DBMS's do not use method syntax to extract an attribute from an object. Rather, the parentheses are dropped, and we shall do so in what follows. For instance, the query of Example 10.22 will be written:

```
SELECT m.year
FROM Movies m
WHERE m.title = 'King Kong';
```

The tuple variable m is still necessary, however.

The dot operator can be used to apply methods as well as to find attribute values within objects. These methods should have the parentheses attached, even if they take no arguments.

Example 10.23: Suppose relation `MovieStar` has been declared to have UDT `StarType`, which we should recall from Example 10.14 has an attribute `address` of type `AddressType`. That type, in turn, has a method `houseNumber()`, which extracts the house number from an object of type `AddressType` (see Example 10.15). Then the query

```
SELECT MAX(s.address.houseNumber())
FROM MovieStar s
```

extracts the address component from a `StarType` object s, then applies the `houseNumber()` method to that `AddressType` object. The result returned is the largest house number of any movie star. □

10.5.3 Generator and Mutator Functions

In order to create data that conforms to a UDT, or to change components of objects with a UDT, we can use two kinds of methods that are created automatically, along with the observer methods, whenever a UDT is defined. These are:

1. A *generator method.* This method has the name of the type and no argument. It may be invoked without being applied to any object. That is, if T is a UDT, then $T()$ returns an object of type T, with no values in its various components.

2. *Mutator methods.* For each attribute x of UDT T, there is a mutator method $x(v)$. When applied to an object of type T, it changes the x attribute of that object to have value v. Notice that the mutator and observer method for an attribute each have the name of the attribute, but differ in that the mutator has an argument.

Example 10.24: We shall write a PSM procedure that takes as arguments a street, a city, and a name, and inserts into the relation `MovieStar` (of type `StarType` according to Example 10.17) an object constructed from these values, using calls to the proper generator and mutator functions. Recall from Example 10.14 that objects of `StarType` have a `name` component that is a character string, but an `address` component that is itself an object of type `AddressType`. The procedure `InsertStar` is shown in Fig. 10.22.

```
 1)  CREATE PROCEDURE InsertStar(
 2)      IN s CHAR(50),
 3)      IN c CHAR(20),
 4)      IN n CHAR(30)
     )
 5)  DECLARE newAddr AddressType;
 6)  DECLARE newStar StarType;

     BEGIN
 7)      SET newAddr = AddressType();
 8)      SET newStar = StarType();
 9)      newAddr.street(s);
10)      newAddr.city(c);
11)      newStar.name(n);
12)      newStar.address(newAddr);
13)      INSERT INTO MovieStar VALUES(newStar);
     END;
```

Figure 10.22: Creating and storing a `StarType` object

Lines (2) through (4) introduce the arguments s, c, and n, which will provide values for a street, city, and star name, respectively. Lines (5) and (6) declare two local variables. Each is of one of the UDT's involved in the type for objects that exist in the relation `MovieStar`. At lines (7) and (8) we create empty objects of each of these two types.

Lines (9) and (10) put real values in the object `newAddr`; these values are taken from the procedure arguments that provide a street and a city. Line (11) similarly installs the argument n as the value of the `name` component in the object `newStar`. Then line (12) takes the entire `newAddr` object and makes it the value of the `address` component in `newStar`. Finally, line (13) inserts the constructed object into relation `MovieStar`. Notice that, as always, a relation that has a UDT as its type has but a single component, even if that component has several attributes, such as `name` and `address` in this example.

To insert a star into `MovieStar`, we can call procedure `InsertStar`.

```
CALL InsertStar('345 Spruce St.', 'Glendale', 'Gwyneth Paltrow');
```

is an example. □

It is much simpler to insert objects into a relation with a UDT if your DBMS provides a generator function that takes values for the attributes of the UDT and returns a suitable object. For example, if we have functions `AddressType(s,c)` and `StarType(n,a)` that return objects of the indicated types, then we can make the insertion at the end of Example 10.24 with an `INSERT` statement of a familiar form:

```
INSERT INTO MovieStar VALUES(
    StarType('Gwyneth Paltrow',
        AddressType('345 Spruce St.', 'Glendale')));
```

10.5.4 Ordering Relationships on UDT's

Objects that are of some UDT are inherently abstract, in the sense that there is no way to compare two objects of the same UDT, either to test whether they are "equal" or whether one is less than another. Even two objects that have all components identical will not be considered equal unless we tell the system to regard them as equal. Similarly, there is no obvious way to sort the tuples of a relation that has a UDT unless we define a function that tells which of two objects of that UDT precedes the other.

Yet there are many SQL operations that require either an equality test or both an equality and a "less than" test. For instance, we cannot eliminate duplicates if we can't tell whether two tuples are equal. We cannot group by an attribute whose type is a UDT unless there is an equality test for that UDT. We cannot use an `ORDER BY` clause or a comparison like $<$ in a `WHERE` clause unless we can compare two elements.

To specify an ordering or comparison, SQL allows us to issue a `CREATE ORDERING` statement for any UDT. There are a number of forms this statement may take, and we shall only consider the two simplest options:

1. The statement

   `CREATE ORDERING FOR` T `EQUALS ONLY BY STATE;`

   says that two members of UDT T are considered equal if all of their corresponding components are equal. There is no $<$ defined on objects of UDT T.

2. The following statement

   `CREATE ORDERING FOR` T
   `ORDERING FULL BY RELATIVE WITH` F`;`

   says that any of the six comparisons (<, <=, >, >=, =, and <>) may be performed on objects of UDT T. To tell how objects x_1 and x_2 compare, we apply the function F to these objects. This function must be written so that $F(x_1, x_2) < 0$ whenever we want to conclude that $x_1 < x_2$; $F(x_1, x_2) = 0$ means that $x_1 = x_2$, and $F(x_1, x_2) > 0$ means that $x_1 > x_2$. If we replace "`ORDERING FULL`" with "`EQUALS ONLY`," then $F(x_1, x_2) = 0$ indicates that $x_1 = x_2$, while any other value of $F(x_1, x_2)$ means that $x_1 \neq x_2$. Comparison by $<$ is impossible in this case.

Example 10.25: Let us consider a possible ordering on the UDT `StarType` from Example 10.14. If we want only an equality on objects of this UDT, we could declare:

```
CREATE ORDERING FOR StarType EQUALS ONLY BY STATE;
```

That statement says that two objects of `StarType` are equal if and only if their names are equal as character strings, and their addresses are equal as objects of UDT `AddressType`.

The problem is that, unless we define an ordering for `AddressType`, an object of that type is not even equal to itself. Thus, we also need to create at least an equality test for `AddressType`. A simple way to do so is to declare that two `AddressType` objects are equal if and only if their streets and cities are each equal as strings. We could do so by:

```
CREATE ORDERING FOR AddressType EQUALS ONLY BY STATE;
```

Alternatively, we could define a complete ordering of `AddressType` objects. One reasonable ordering is to order addresses first by cities, alphabetically, and among addresses in the same city, by street address, alphabetically. To do so, we have to define a function, say `AddrLEG`, that takes two `AddressType` arguments and returns a negative, zero, or positive value to indicate that the first is less than, equal to, or greater than the second. We declare:

```
CREATE ORDERING FOR AddressType
ORDERING FULL BY RELATIVE WITH AddrLEG;
```

The function `AddrLEG` is shown in Fig. 10.23. Notice that if we reach line (7), it must be that the two `city` components are the same, so we compare the `street` components. Likewise, if we reach line (9), the only remaining possibility is that the cities are the same and the first street precedes the second alphabetically. □

```
1)  CREATE FUNCTION AddrLEG(
2)      x1 AddressType,
3)      x2 AddressType
4)  ) RETURNS INTEGER

5)  IF x1.city() < x2.city() THEN RETURN(-1)
6)  ELSEIF x1.city() > x2.city() THEN RETURN(1)
7)  ELSEIF x1.street() < x2.street() THEN RETURN(-1)
8)  ELSEIF x1.street() = x2.street() THEN RETURN(0)
9)  ELSE RETURN(1)
    END IF;
```

Figure 10.23: A comparison function for address objects

In practice, commercial DBMS's each have their own way of allowing the user to define comparisons for a UDT. In addition to the two approaches mentioned above, some of the capabilities offered are:

a) *Strict Object Equality.* Two objects are equal if and only if they are the same object.

b) *Method-Defined Equality.* A function is applied to two objects and returns true or false, depending on whether or not the two objects should be considered equal.

c) *Method-Defined Mapping.* A function is applied to one object and returns a real number. Objects are compared by comparing the real numbers returned.

10.5.5 Exercises for Section 10.5

Exercise 10.5.1: Use the `StarsIn` relation of Example 10.20 and the `Movies` and `MovieStar` relations accessible through `StarsIn` to write the following queries:

a) Find the names of the stars of *Dogma*.

! b) Find the titles and years of all movies in which at least one star lives in Malibu.

c) Find all the movies (objects of type `MovieType`) that starred Melanie Griffith.

! d) Find the movies (title and year) with at least five stars.

Exercise 10.5.2: Using your schema from Exercise 10.4.3, write the following queries. Don't forget to use references whenever appropriate.

a) Find the manufacturers of PC's with a hard disk larger than 60 gigabytes.

b) Find the manufacturers of laser printers.

! c) Produce a table giving for each model of laptop, the model of the laptop having the highest processor speed of any laptop made by the same manufacturer.

Exercise 10.5.3: Using your schema from Exercise 10.4.5, write the following queries. Don't forget to use references whenever appropriate and avoid joins (i.e., subqueries or more than one tuple variable in the `FROM` clause).

a) Find the ships with a displacement of more than 35,000 tons.

b) Find the battles in which at least one ship was sunk.

! c) Find the classes that had ships launched after 1930.

!! d) Find the battles in which at least one US ship was damaged.

Exercise 10.5.4: Assuming the function `AddrLEG` of Fig. 10.23 is available, write a suitable function to compare objects of type `StarType`, and declare your function to be the basis of the ordering of `StarType` objects.

! **Exercise 10.5.5:** Write a procedure to take a star name as argument and delete from `StarsIn` and `MovieStar` all tuples involving that star.

10.6 On-Line Analytic Processing

An important application of databases is examination of data for patterns or trends. This activity, called *OLAP* (standing for *On-Line Analytic Processing* and pronounced "oh-lap"), generally involves highly complex queries that use one or more aggregations. These queries are often termed *OLAP queries* or *decision-support queries*. Some examples will be given in Section 10.6.2. A typical example is for a company to search for those of its products that have markedly increasing or decreasing overall sales.

Decision-support queries typically examine very large amounts of data, even if the query results are small. In contrast, common database operations, such

as bank deposits or airline reservations, each touch only a tiny portion of the database; the latter type of operation is often referred to as *OLTP* (*On-Line Transaction Processing*, spoken "oh-ell-tee-pee").

A recent trend in DBMS's is to provide specialized support for OLAP queries. For example, systems often support a "data cube" in some way. We shall discuss the architecture of these systems in Section 10.7.

10.6.1 OLAP and Data Warehouses

It is common for OLAP applications to take place in a separate copy of the master database, called a *data warehouse*. Data from many separate databases may be integrated into the warehouse. In a common scenario, the warehouse is only updated overnight, while the analysts work on a frozen copy during the day. The warehouse data thus gets out of date by as much as 24 hours, which limits the timeliness of its answers to OLAP queries, but the delay is tolerable in many decision-support applications.

There are several reasons why data warehouses play an important role in OLAP applications. First, the warehouse may be necessary to organize and centralize data in a way that supports OLAP queries; the data may initially be scattered across many different databases. But often more important is the fact that OLAP queries, being complex and touching much of the data, take too much time to be executed in a transaction-processing system with high throughput requirements. Recall the discussion of serializable transactions in Section 6.6. Trying to run a long transaction that needed to touch much of the database serializably with other transactions would stall ordinary OLTP operations more than could be tolerated. For instance, recording new sales as they occur might not be permitted if there were a concurrent OLAP query computing average sales.

10.6.2 OLAP Applications

A common OLAP application uses a warehouse of sales data. Major store chains will accumulate terabytes of information representing every sale of every item at every store. Queries that aggregate sales into groups and identify significant groups can be of great use to the company in predicting future problems and opportunities.

Example 10.26: Suppose the Aardvark Automobile Co. builds a data warehouse to analyze sales of its cars. The schema for the warehouse might be:

```
Sales(serialNo, date, dealer, price)
Autos(serialNo, model, color)
Dealers(name, city, state, phone)
```

A typical decision-support query might examine sales on or after April 1, 2006 to see how the recent average price per vehicle varies by state. Such a query is shown in Fig. 10.24.

```
SELECT state, AVG(price)
FROM Sales, Dealers
WHERE Sales.dealer = Dealers.name AND
      date >= '2006-01-04'
GROUP BY state;
```

Figure 10.24: Find average sales price by state

Notice how the query of Fig. 10.24 touches much of the data of the database, as it classifies every recent `Sales` fact by the state of the dealer that sold it. In contrast, a typical OLTP query such as "find the price at which the auto with serial number 123 was sold," would touch only a single tuple of the data, provided there was an index on serial number. □

For another OLAP example, consider a credit-card company trying to decide whether applicants for a card are likely to be credit-worthy. The company creates a warehouse of all its current customers and their payment history. OLAP queries search for factors, such as age, income, home-ownership, and zip-code, that might help predict whether customers will pay their bills on time. Similarly, hospitals may use a warehouse of patient data — their admissions, tests administered, outcomes, diagnoses, treatments, and so on — to analyze for risks and select the best modes of treatment.

10.6.3 A Multidimensional View of OLAP Data

In typical OLAP applications there is a central relation or collection of data, called the *fact table*. A fact table represents events or objects of interest, such as sales in Example 10.26. Often, it helps to think of the objects in the fact table as arranged in a multidimensional space, or "cube." Figure 10.25 suggests three-dimensional data, represented by points within the cube; we have called the dimensions car, dealer, and date, to correspond to our earlier example of automobile sales. Thus, in Fig. 10.25 we could think of each point as a sale of a single automobile, while the dimensions represent properties of that sale.

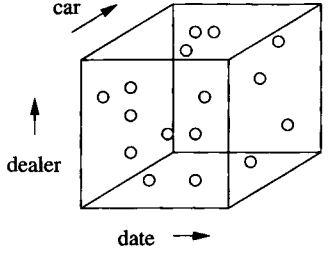


Figure 10.25: Data organized in a multidimensional space

A data space such as Fig. 10.25 will be referred to informally as a "data cube," or more precisely as a *raw-data cube* when we want to distinguish it from the more complex "data cube" of Section 10.7. The latter, which we shall refer to as a *formal* data cube when a distinction from the raw-data cube is needed, differs from the raw-data cube in two ways:

1. It includes aggregations of the data in all subsets of dimensions, as well as the data itself.

2. Points in the formal data cube may represent an initial aggregation of points in the raw-data cube. For instance, instead of the "car" dimension representing each individual car (as we suggested for the raw-data cube), that dimension might be aggregated by model only. There are points of a formal data cube that represent the total sales of all cars of a given model by a given dealer on a given day.

The distinctions between the raw-data cube and the formal data cube are reflected in the two broad directions that have been taken by specialized systems that support cube-structured data for OLAP:

1. *ROLAP*, or *Relational OLAP*. In this approach, data may be stored in relations with a specialized structure called a "star schema," described in Section 10.6.4. One of these relations is the "fact table," which contains the *raw*, or unaggregated, data, and corresponds to what we called the raw-data cube. Other relations give information about the values along each dimension. The query language, index structures, and other capabilities of the system may be tailored to the assumption that data is organized this way.

2. *MOLAP*, or *Multidimensional OLAP*. Here, a specialized structure, the formal "data cube" mentioned above, is used to hold the data, including its aggregates. Nonrelational operators may be implemented by the system to support OLAP queries on data in this structure.

10.6.4 Star Schemas

A *star schema* consists of the schema for the fact table, which links to several other relations, called "dimension tables." The fact table is at the center of the "star," whose points are the dimension tables. A fact table normally has several attributes that represent *dimensions*, and one or more *dependent* attributes that represent properties of interest for the point as a whole. For instance, dimensions for sales data might include the date of the sale, the place (store) of the sale, the type of item sold, the method of payment (e.g., cash or a credit card), and so on. The dependent attribute(s) might be the sales price, the cost of the item, or the tax, for instance.

Example 10.27: The `Sales` relation from Example 10.26

```
Sales(serialNo, date, dealer, price)
```

is a fact table. The dimensions are:

1. `serialNo`, representing the automobile sold, i.e., the position of the point in the space of possible automobiles.
2. `date`, representing the day of the sale, i.e., the position of the event in the time dimension.
3. `dealer`, representing the position of the event in the space of possible dealers.

The one dependent attribute is `price`, which is what OLAP queries to this database will typically request in an aggregation. However, queries asking for a count, rather than sum or average price would also make sense, e.g., "list the total number of sales for each dealer in the month of May, 2006." □

Supplementing the fact table are *dimension tables* describing the values along each dimension. Typically, each dimension attribute of the fact table is a foreign key, referencing the key of the corresponding dimension table, as suggested by Fig. 10.26. The attributes of the dimension tables also describe the possible groupings that would make sense in a SQL `GROUP BY` query. An example should make the ideas clearer.

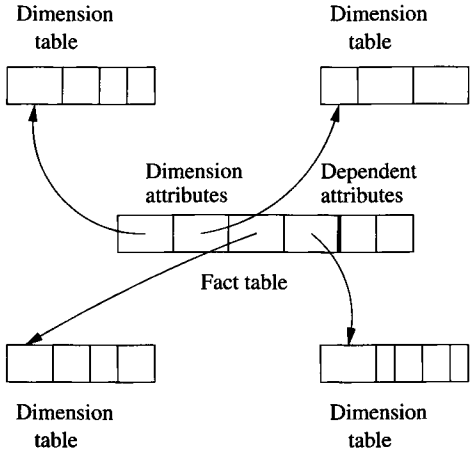


Figure 10.26: The dimension attributes in the fact table reference the keys of the dimension tables

Example 10.28: For the automobile data of Example 10.26, two of the three dimension tables might be:

```
Autos(serialNo, model, color)
Dealers(name, city, state, phone)
```

Attribute serialNo in the fact table Sales is a foreign key, referencing serialNo of dimension table Autos.[2] The attributes Autos.model and Autos.color give properties of a given auto. If we join the fact table Sales with the dimension table Autos, then the attributes model and color may be used for grouping sales in interesting ways. For instance, we can ask for a breakdown of sales by color, or a breakdown of sales of the Gobi model by month and dealer.

Similarly, attribute dealer of Sales is a foreign key, referencing name of the dimension table Dealers. If Sales and Dealers are joined, then we have additional options for grouping our data; e.g., we can ask for a breakdown of sales by state or by city, as well as by dealer.

One might wonder where the dimension table for time (the date attribute of Sales) is. Since time is a physical property, it does not make sense to store facts about time in a database, since we cannot change the answer to questions such as "in what year does the day July 5, 2007 appear?" However, since grouping by various time units, such as weeks, months, quarters, and years, is frequently desired by analysts, it helps to build into the database a notion of time, as if there were a time "dimension table" such as

```
Days(day, week, month, year)
```

A typical tuple of this imaginary "relation" would be $(5, 27, 7, 2007)$, representing July 5, 2007. The interpretation is that this day is the fifth day of the seventh month of the year 2007; it also happens to fall in the 27th full week of the year 2007. There is a certain amount of redundancy, since the week is calculable from the other three attributes. However, weeks are not exactly commensurate with months, so we cannot obtain a grouping by months from a grouping by weeks, or vice versa. Thus, it makes sense to imagine that both weeks and months are represented in this "dimension table." □

10.6.5 Slicing and Dicing

We can think of the points of the raw-data cube as partitioned along each dimension at some level of granularity. For example, in the time dimension, we might partition ("group by" in SQL terms) according to days, weeks, months, years, or not partition at all. For the cars dimension, we might partition by model, by color, by both model and color, or not partition. For dealers, we can partition by dealer, by city, by state, or not partition.

A choice of partition for each dimension "dices" the cube, as suggested by Fig. 10.27. The result is that the cube is divided into smaller cubes that represent groups of points whose statistics are aggregated by a query that performs this partitioning in its GROUP BY clause. Through the WHERE clause, a query also

[2]It happens that serialNo is also a key for the Sales relation, but there need not be an attribute that is both a key for the fact table and a foreign key for some dimension table.

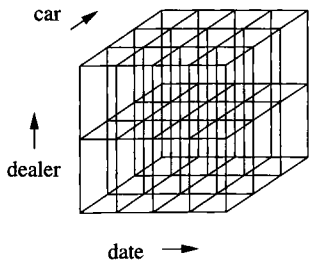


Figure 10.27: Dicing the cube by partitioning along each dimension

has the option of focusing on particular partitions along one or more dimensions (i.e., on a particular "slice" of the cube).

Example 10.29: Figure 10.28 suggests a query in which we ask for a slice in one dimension (the date), and dice in two other dimensions (car and dealer). The date is divided into four groups, perhaps the four years over which data has been accumulated. The shading in the diagram suggests that we are only interested in one of these years.

The cars are partitioned into three groups, perhaps sedans, SUV's, and convertibles, while the dealers are partitioned into two groups, perhaps the eastern and western regions. The result of the query is a table giving the total sales in six categories for the one year of interest. □

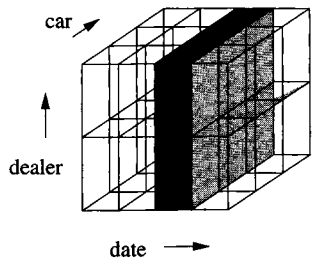


Figure 10.28: Selecting a slice of a diced cube

The general form of a so-called "slicing and dicing" query is thus:

```
SELECT <grouping attributes and aggregations>
FROM <fact table joined with some dimension tables>
WHERE <certain attributes are constant>
GROUP BY <grouping attributes>;
```

Example 10.30: Let us continue with our automobile example, but include the conceptual `Days` dimension table for time discussed in Example 10.28. If

Drill-Down and Roll-Up

Example 10.30 illustrates two common patterns in sequences of queries that slice-and-dice the data cube.

1. *Drill-down* is the process of partitioning more finely and/or focusing on specific values in certain dimensions. Each of the steps except the last in Example 10.30 is an instance of drill-down.
2. *Roll-up* is the process of partitioning more coarsely. The last step, where we grouped by years instead of months to eliminate the effect of randomness in the data, is an example of roll-up.

the Gobi isn't selling as well as we thought it would, we might try to find out which colors are not doing well. This query uses only the `Autos` dimension table and can be written in SQL as:

```
SELECT color, SUM(price)
FROM Sales NATURAL JOIN Autos
WHERE model = 'Gobi'
GROUP BY color;
```

This query dices by color and then slices by model, focusing on a particular model, the Gobi, and ignoring other data.

Suppose the query doesn't tell us much; each color produces about the same revenue. Since the query does not partition on time, we only see the total over all time for each color. We might suppose that the recent trend is for one or more colors to have weak sales. We may thus issue a revised query that also partitions time by month. This query is:

```
SELECT color, month, SUM(price)
FROM (Sales NATURAL JOIN Autos) JOIN Days ON date = day
WHERE model = 'Gobi'
GROUP BY color, month;
```

It is important to remember that the `Days` relation is not a conventional stored relation, although we may treat it as if it had the schema

```
Days(day, week, month, year)
```

The ability to use such a "relation" is one way that a system specialized to OLAP queries could differ from a conventional DBMS.

We might discover that red Gobis have not sold well recently. The next question we might ask is whether this problem exists at all dealers, or whether

only some dealers have had low sales of red Gobis. Thus, we further focus the query by looking at only red Gobis, and we partition along the dealer dimension as well. This query is:

```
SELECT dealer, month, SUM(price)
FROM (Sales NATURAL JOIN Autos) JOIN Days ON date = day
WHERE model = 'Gobi' AND color = 'red'
GROUP BY month, dealer;
```

At this point, we find that the sales per month for red Gobis are so small that we cannot observe any trends easily. Thus, we decide that it was a mistake to partition by month. A better idea would be to partition only by years, and look at only the last two years (2006 and 2007, in this hypothetical example). The final query is shown in Fig. 10.29. □

```
SELECT dealer, year, SUM(price)
FROM (Sales NATURAL JOIN Autos) JOIN Days ON date = day
WHERE model = 'Gobi' AND
      color = 'red' AND
      (year = 2006 OR year = 2007)
GROUP BY year, dealer;
```

Figure 10.29: Final slicing-and-dicing query about red Gobi sales

10.6.6 Exercises for Section 10.6

Exercise 10.6.1: An on-line seller of computers wishes to maintain data about orders. Customers can order their PC with any of several processors, a selected amount of main memory, any of several disk units, and any of several CD or DVD readers. The fact table for such a database might be:

```
Orders(cust, date, proc, memory, hd, od, quant, price)
```

We should understand attribute `cust` to be an ID that is the foreign key for a dimension table about customers, and understand attributes `proc`, `hd` (hard disk), and `od` (optical disk: CD or DVD, typically) analogously. For example, an `hd` ID might be elaborated in a dimension table giving the manufacturer of the disk and several disk characteristics. The `memory` attribute is simply an integer: the number of megabytes of memory ordered. The `quant` attribute is the number of machines of this type ordered by this customer, and the `price` attribute is the total cost of each machine ordered.

a) Which are dimension attributes, and which are dependent attributes?

b) For some of the dimension attributes, a dimension table is likely to be needed. Suggest appropriate schemas for these dimension tables.

! **Exercise 10.6.2:** Suppose that we want to examine the data of Exercise 10.6.1 to find trends and thus predict which components the company should order more of. Describe a series of drill-down and roll-up queries that could lead to the conclusion that customers are beginning to prefer a DVD drive to a CD drive.

10.7 Data Cubes

In this section, we shall consider the "formal" data cube and special operations on data presented in this form. Recall from Section 10.6.3 that the formal data cube (just "data cube" in this section) precomputes all possible aggregates in a systematic way. Surprisingly, the amount of extra storage needed is often tolerable, and as long as the warehoused data does not change, there is no penalty incurred trying to keep all the aggregates up-to-date.

In the data cube, it is normal for there to be some aggregation of the raw data of the fact table before it is entered into the data-cube and its further aggregates computed. For instance, in our cars example, the dimension we thought of as a serial number in the star schema might be replaced by the model of the car. Then, each point of the data cube becomes a description of a model, a dealer and a date, together with the sum of the sales for that model, on that date, by that dealer. We shall continue to call the points of the (formal) data cube a "fact table," even though the interpretation of the points may be slightly different from fact tables in a star schema built from a raw-data cube.

10.7.1 The Cube Operator

Given a fact table F, we can define an augmented table CUBE(F) that adds an additional value, denoted *, to each dimension. The * has the intuitive meaning "any," and it represents aggregation along the dimension in which it appears. Figure 10.30 suggests the process of adding a border to the cube in each dimension, to represent the * value and the aggregated values that it implies. In this figure we see three dimensions, with the lightest shading representing aggregates in one dimension, darker shading for aggregates over two dimensions, and the darkest cube in the corner for aggregation over all three dimensions. Notice that if the number of values along each dimension is reasonably large, then the "border" represents only a small addition to the volume of the cube (i.e., the number of tuples in the fact table). In that case, the size of the stored data CUBE(F) is not much greater than the size of F itself.

A tuple of the table CUBE(F) that has * in one or more dimensions will have for each dependent attribute the sum (or another aggregate function) of the values of that attribute in all the tuples that we can obtain by replacing

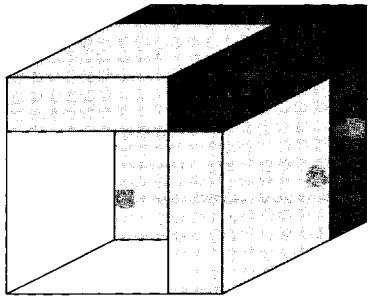


Figure 10.30: The cube operator augments a data cube with a border of aggregations in all combinations of dimensions

the *'s by real values. In effect, we build into the data the result of aggregating along any set of dimensions. Notice, however, that the CUBE operator does not support aggregation at intermediate levels of granularity based on values in the dimension tables. For instance, we may either leave data broken down by day (or whatever the finest granularity for time is), or we may aggregate time completely, but we cannot, with the CUBE operator alone, aggregate by weeks, months, or years.

Example 10.31: Let us reconsider the Aardvark database from Example 10.26 in the light of what the CUBE operator can give us. Recall the fact table from that example is

```
Sales(serialNo, date, dealer, price)
```

However, the dimension represented by `serialNo` is not well suited for the cube, since the serial number is a key for `Sales`. Thus, summing the price over all dates, or over all dealers, but keeping the serial number fixed has no effect; we would still get the "sum" for the one auto with that serial number. A more useful data cube would replace the serial number by the two attributes — model and color — to which the serial number connects `Sales` via the dimension table `Autos`. Notice that if we replace `serialNo` by `model` and `color`, then the cube no longer has a key among its dimensions. Thus, an entry of the cube would have the total sales price for all automobiles of a given model, with a given color, by a given dealer, on a given date.

There is another change that is useful for the data-cube implementation of the `Sales` fact table. Since the CUBE operator normally sums dependent variables, and we might want to get average prices for sales in some category, we need both the sum of the prices for each category of automobiles (a given model of a given color sold on a given day by a given dealer) and the total number of sales in that category. Thus, the relation `Sales` to which we apply the CUBE operator is

```
Sales(model, color, date, dealer, val, cnt)
```

The attribute `val` is intended to be the total price of all automobiles for the given model, color, date, and dealer, while `cnt` is the total number of automobiles in that category.

Now, let us consider the relation CUBE(`Sales`). A hypothetical tuple that would be in CUBE(`Sales`) is:

```
('Gobi', 'red', '2001-05-21', 'Friendly Fred', 45000, 2)
```

The interpretation is that on May 21, 2001, dealer Friendly Fred sold two red Gobis for a total of $45,000. In `Sales`, this tuple might appear as well, or there could be in `Sales` two tuples, each with a `cnt` of 1, whose `val`'s summed to 45,000.

The tuple

```
('Gobi', *, '2001-05-21', 'Friendly Fred', 152000, 7)
```

says that on May 21, 2001, Friendly Fred sold seven Gobis of all colors, for a total price of $152,000. Note that this tuple is in CUBE(`Sales`) but not in `Sales`.

Relation CUBE(`Sales`) also contains tuples that represent the aggregation over more than one attribute. For instance,

```
('Gobi', *, '2001-05-21', *, 2348000, 100)
```

says that on May 21, 2001, there were 100 Gobis sold by all the dealers, and the total price of those Gobis was $2,348,000.

```
('Gobi', *, *, *, 1339800000, 58000)
```

Says that over all time, dealers, and colors, 58,000 Gobis have been sold for a total price of $1,339,800,000. Lastly, the tuple

```
(*, *, *, *, 3521727000, 198000)
```

tells us that total sales of all Aardvark models in all colors, over all time at all dealers is 198,000 cars for a total price of $3,521,727,000. □

10.7.2 The Cube Operator in SQL

SQL gives us a way to apply the cube operator within queries. If we add the term `WITH CUBE` to a group-by clause, then we get not only the tuple for each group, but also the tuples that represent aggregation along one or more of the dimensions along which we have grouped. These tuples appear in the result with `NULL` where we have used *.

Example 10.32: We can construct a materialized view that is the data cube we called CUBE(Sales) in Example 10.31 by the following:

```
CREATE MATERIALIZED VIEW SalesCube AS
    SELECT model, color, date, dealer, SUM(val), SUM(cnt)
    FROM Sales
    GROUP BY model, color, date, dealer WITH CUBE;
```

The view SalesCube will then contain not only the tuples that are implied by the group-by operation, such as

```
('Gobi', 'red', '2001-05-21', 'Friendly Fred', 45000, 2)
```

but will also contain those tuples of CUBE(Sales) that are constructed by rolling up the dimensions listed in the GROUP BY. Some examples of such tuples would be:

```
('Gobi', NULL, '2001-05-21', 'Friendly Fred', 152000, 7)
('Gobi', NULL, '2001-05-21', NULL, 2348000, 100)
('Gobi', NULL, NULL, NULL, 1339800000, 58000)
(NULL, NULL, NULL, NULL, 3521727000, 198000)
```

Recall that NULL is used to indicate a rolled-up dimension, equivalent to the * we used in the abstract CUBE operator's result. □

A variant of the CUBE operator, called ROLLUP, produces the additional aggregated tuples only if they aggregate over a tail of the sequence of grouping attributes. We indicate this option by appending WITH ROLLUP to the group-by clause.

Example 10.33: We can get the part of the data cube for Sales that is constructed by the ROLLUP operator with:

```
CREATE MATERIALIZED VIEW SalesRollup AS
    SELECT model, color, date, dealer, SUM(val), SUM(cnt)
    FROM Sales
    GROUP BY model, color, date, dealer WITH ROLLUP;
```

The view SalesRollup will contain tuples

```
('Gobi', 'red', '2001-05-21', 'Friendly Fred', 45000, 2)
('Gobi', 'red', '2001-05-21', NULL, 3678000, 135)
('Gobi', 'red', NULL, NULL, 657100000, 34566)
('Gobi', NULL, NULL, NULL, 1339800000, 58000)
(NULL, NULL, NULL, NULL, 3521727000, 198000)
```

because these tuples represent aggregation along some dimension and all dimensions, if any, that follow it in the list of grouping attributes.

However, SalesRollup would not contain tuples such as

```
('Gobi', NULL, '2001-05-21', 'Friendly Fred', 152000, 7)
('Gobi', NULL, '2001-05-21', NULL, 2348000, 100)
```

These each have NULL in a dimension (color in both cases) but do not have NULL in one or more of the following dimension attributes. □

10.7.3 Exercises for Section 10.7

Exercise 10.7.1: What is the ratio of the size of CUBE(F) to the size of F if fact table F has the following characteristics?

a) F has ten dimension attributes, each with ten different values.

b) F has ten dimension attributes, each with two different values.

Exercise 10.7.2: Use the materialized view SalesCube from Example 10.32 to answer the following queries:

a) Find the total sales of blue cars for each dealer.

b) Find the total number of green Gobis sold by dealer "Smilin' Sally."

c) Find the average number of Gobis sold on each day of March, 2007 by each dealer.

! **Exercise 10.7.3:** What help, if any, would the rollup SalesRollup of Example 10.33 be for each of the queries of Exercise 10.7.2?

Exercise 10.7.4: In Exercise 10.6.1 we spoke of PC-order data organized as a fact table with dimension tables for attributes cust, proc, memory, hd, and od. That is, each tuple of the fact table Orders has an ID for each of these attributes, leading to information about the PC involved in the order. Write a SQL query that will produce the data cube for this fact table.

Exercise 10.7.5: Answer the following queries using the data cube from Exercise 10.7.4. If necessary, use dimension tables as well. You may invent suitable names and attributes for the dimension tables.

a) Find, for each processor speed, the total number of computers ordered in each month of the year 2007.

b) List for each type of hard disk (e.g., SCSI or IDE) and each processor type the number of computers ordered.

c) Find the average price of computers with 3.0 gigahertz processors for each month from Jan., 2005.

! **Exercise 10.7.6:** The cube tuples mentioned in Example 10.32 are not in the rollup of Example 10.33. Are there other rollups that would contain these tuples?

!! **Exercise 10.7.7:** If the fact table F to which we apply the CUBE operator is sparse (i.e., there are many fewer tuples in F than the product of the number of possible values along each dimension), then the ratio of the sizes of CUBE(F) and F can be very large. How large can it be?

10.8 Summary of Chapter 10

- ✦ *Privileges*: For security purposes, SQL systems allow many different kinds of privileges to be managed for database elements. These privileges include the right to select (read), insert, delete, or update relations, the right to reference relations (refer to them in a constraint), and the right to create triggers.

- ✦ *Grant Diagrams*: Privileges may be granted by owners to other users or to the general user PUBLIC. If granted with the grant option, then these privileges may be passed on to others. Privileges may also be revoked. The grant diagram is a useful way to remember enough about the history of grants and revocations to keep track of who has what privilege and from whom they obtained those privileges.

- ✦ *SQL Recursive Queries*: In SQL, one can define a relation recursively — that is, in terms of itself. Or, several relations can be defined to be mutually recursive.

- ✦ *Monotonicity*: Negations and aggregations involved in a SQL recursion must be monotone — inserting tuples in one relation does not cause tuples to be deleted from any relation, including itself. Intuitively, a relation may not be defined, directly or indirectly, in terms of a negation or aggregation of itself.

- ✦ *The Object-Relational Model*: An alternative to pure object-oriented database models like ODL is to extend the relational model to include the major features of object-orientation. These extensions include nested relations, i.e., complex types for attributes of a relation, including relations as types. Other extensions include methods defined for these types, and the ability of one tuple to refer to another through a reference type.

- ✦ *User-Defined Types in SQL*: Object-relational capabilities of SQL are centered around the UDT, or user-defined type. These types may be declared by listing their attributes and other information, as in table declarations. In addition, methods may be declared for UDT's.

- ✦ *Relations With a UDT as Type*: Instead of declaring the attributes of a relation, we may declare that relation to have a UDT. If we do so, then its tuples have one component, and this component is an object of the UDT.

- *Reference Types*: A type of an attribute can be a reference to a UDT. Such attributes essentially are pointers to objects of that UDT.

- *Object Identity for UDT's*: When we create a relation whose type is a UDT, we declare an attribute to serve as the "object-ID" of each tuple. This component is a reference to the tuple itself. Unlike in object-oriented systems, this "OID" column may be accessed by the user, although it is rarely meaningful.

- *Accessing components of a UDT*: SQL provides observer and mutator functions for each attribute of a UDT. These functions, respectively, return and change the value of that attribute when applied to any object of that UDT.

- *Ordering Functions for UDT's*: In order to compare objects, or to use SQL operations such as `DISTINCT`, `GROUP BY`, or `ORDER BY`, it is necessary for the implementer of a UDT to provide a function that tells whether two objects are equal or whether one precedes the other.

- *OLAP*: On-line analytic processing involves complex queries that touch all or much of the data, at the same time. Often, a separate database, called a data warehouse, is constructed to run such queries while the actual database is used for short-term transactions (OLTP, or on-line transaction processing).

- *ROLAP and MOLAP*: It is frequently useful, for OLAP queries, to think of the data as residing in a multidimensional space, with dimensions corresponding to independent aspects of the data represented. Systems that support such a view of data take either a relational point of view (ROLAP, or relational-OLAP systems), or use the specialized data-cube model (MOLAP, or multidimensional-OLAP systems).

- *Star Schemas*: In a star schema, each data element (e.g., a sale of an item) is represented in one relation, called the fact table, while information helping to interpret the values along each dimension (e.g., what kind of product is item 1234?) is stored in a dimension table for each dimension.

- *The Cube Operator*: A specialized operator called cube pre-aggregates the fact table along all subsets of dimensions. It may add little to the space needed by the fact table, and greatly increases the speed with which many OLAP queries can be answered.

- *Data Cubes in SQL*: We can turn the result of a query into a data cube by appending `WITH CUBE` to a group-by clause. We can also construct a portion of the cube by using `WITH ROLLUP` there.

10.9 References for Chapter 10

The ideas behind the SQL authorization mechanism originated in [4] and [1].

Material on object-relational features of SQL can be obtained as described in the bibliographic notes to Chapter 6.

The source of the SQL-99 proposal for recursion is [2]. This proposal, and its monotonicity requirement, built on foundations developed over many years, involving recursion and negation in Datalog; see [5].

The cube operator was proposed in [3].

1. R. Fagin, "On an authorization mechanism," *ACM Transactions on Database Systems* **3**:3, pp. 310–319, 1978.

2. S. J. Finkelstein, N. Mattos, I. S. Mumick, and H. Pirahesh, "Expressing recursive queries in SQL," ISO WG3 report X3H2–96–075, March, 1996.

3. J. N. Gray, A. Bosworth, A. Layman, and H. Pirahesh, "Data cube: a relational aggregation operator generalizing group-by, cross-tab, and sub-totals," *Proc. Intl. Conf. on Data Engineering* (1996), pp. 152–159.

4. P. P. Griffiths and B. W. Wade, "An authorization mechanism for a relational database system," *ACM Transactions on Database Systems* **1**:3, pp. 242–255, 1976.

5. J. D. Ullman, *Principles of Database and Knowledge-Base Systems, Volume I*, Computer Science Press, New York, 1988.

Part III

Modeling and Programming for Semistructured Data